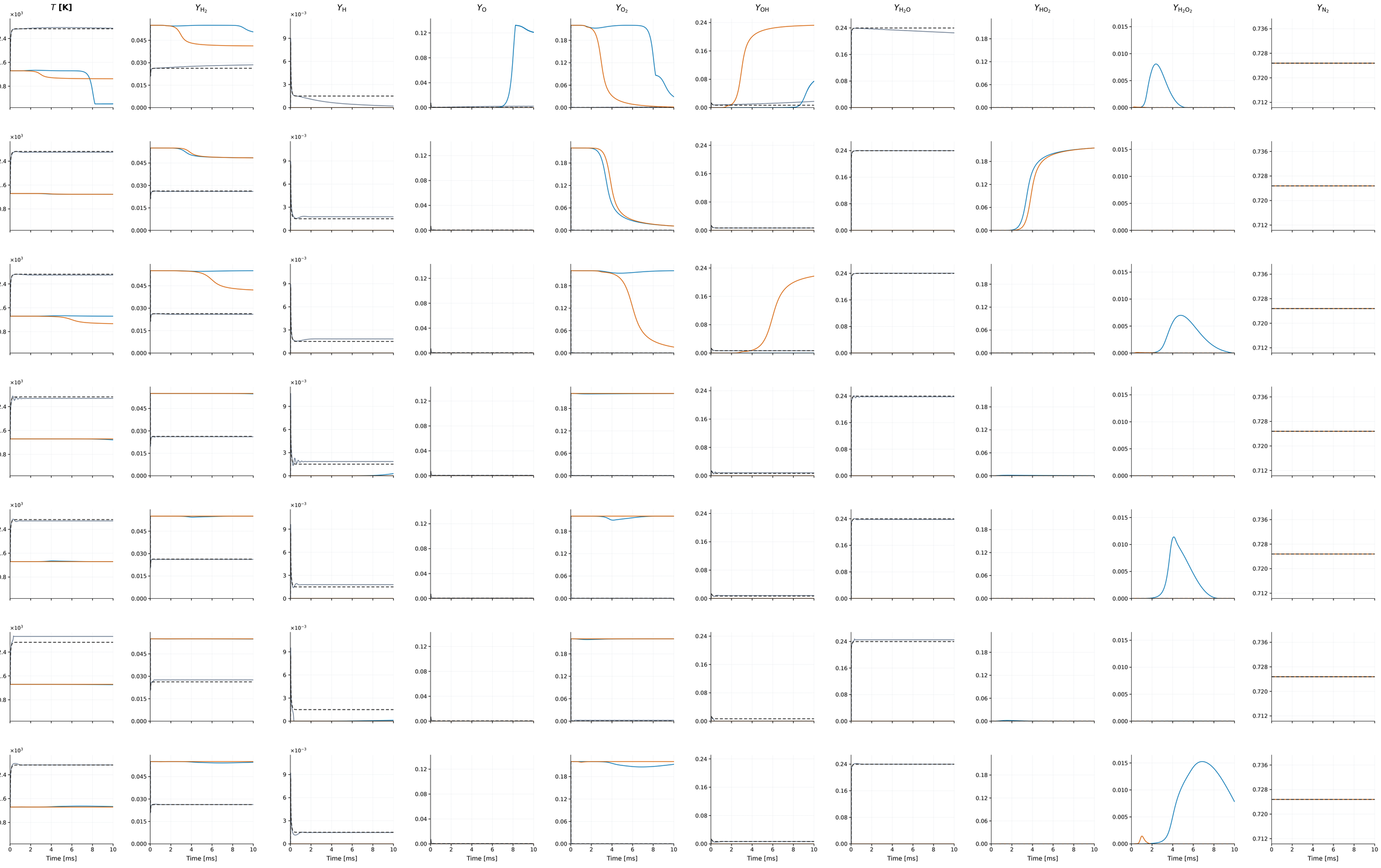
# Lambda 1000 continuation: Cantera vs NN | Test IC 04

$T_0$ = 1317.5 K  $P_0$ = 1.244 atm  $\phi_0$ = 1.984  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=1000) · Soft long (lambda=1000)




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 04 | Cantera vs NN NRMSE

T0=1317.5 K  P0=1.244 atm  phi0=1.984  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

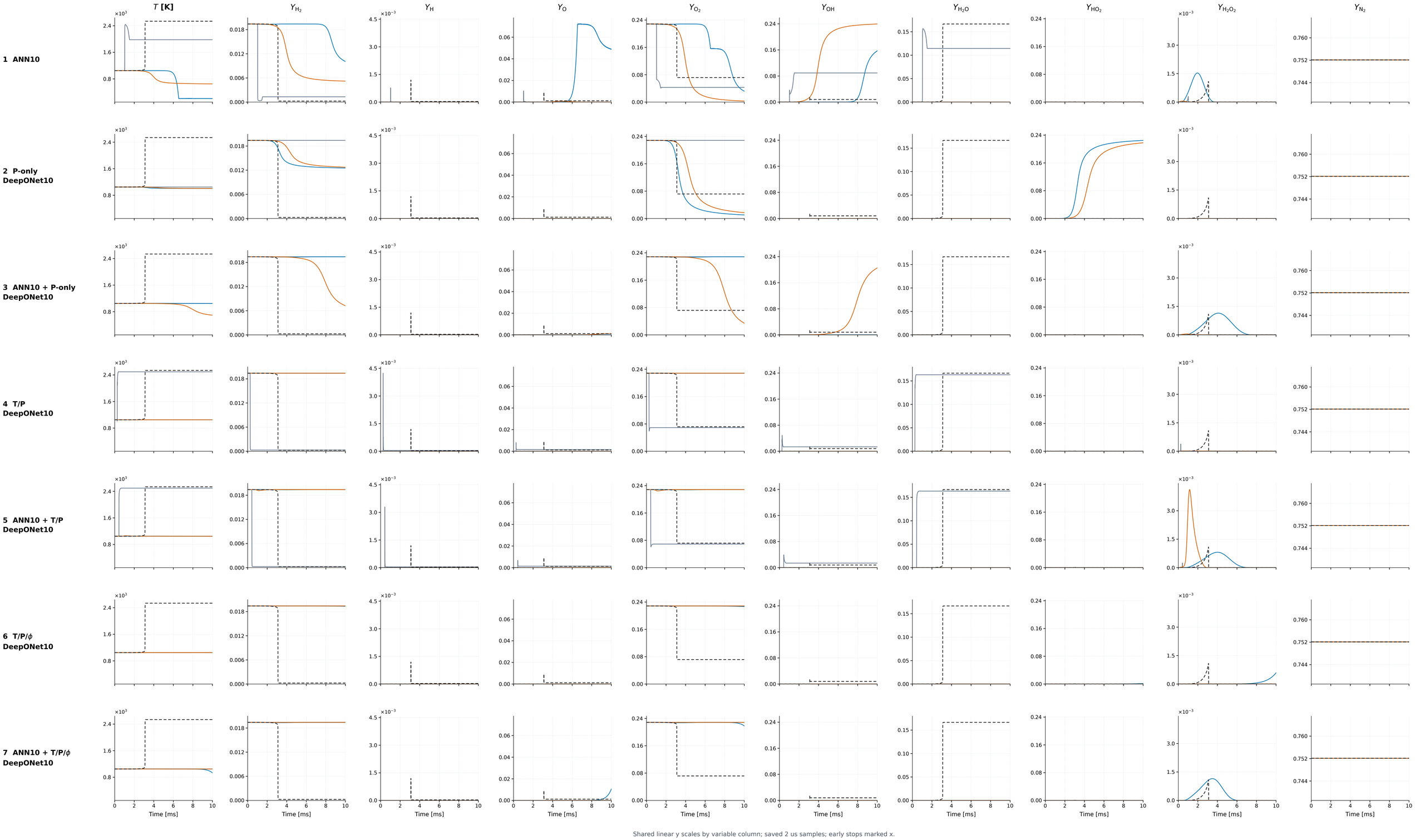
| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 0.0108 | 0.0583 | 0.593 | 2.33 | 0.0953 | 0.832 | 0.0354 | 0.149 | 0.418 | 0 | Complete |
| **1 ANN10** | soft10k | 0.623 | 1.07 | 1 | 100 | 18.8 | 2.47 | 1 | 0.998 | 2.38e+04 | 0 | Complete |
| **1 ANN10** | long | 0.584 | 0.767 | 1 | 1 | 11.1 | 25.9 | 1 | 0.998 | 232 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 8.74e-03 | 0.0113 | 0.165 | 0.217 | 0.0395 | 0.0155 | 2.22e-03 | 0.206 | 0.266 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.522 | 0.942 | 1 | 1 | 12.3 | 1 | 1 | 4.93e+04 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.522 | 0.952 | 1 | 1 | 13 | 1 | 1 | 4.77e+04 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 9.87e-03 | 0.0137 | 0.196 | 0.258 | 0.0434 | 0.031 | 2.92e-03 | 0.155 | 0.2 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.514 | 1.08 | 1 | 1 | 21 | 0.999 | 1 | 0.998 | 2.71e+04 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.553 | 0.923 | 1 | 1 | 15.9 | 17.1 | 1 | 0.998 | 277 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 0.0189 | 0.0104 | 0.227 | 0.392 | 0.0858 | 0.218 | 9.30e-03 | 0.336 | 0.28 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.517 | 1.09 | 0.982 | 1 | 21.2 | 1 | 1 | 268 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.516 | 1.09 | 1 | 1 | 21.2 | 1 | 1 | 0.992 | 2.36 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 0.016 | 8.47e-03 | 0.186 | 0.267 | 0.0804 | 0.21 | 8.43e-03 | 0.233 | 0.181 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.514 | 1.08 | 1 | 0.998 | 21 | 0.999 | 1 | 2.72 | 3.47e+04 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.516 | 1.09 | 1 | 1 | 21.2 | 1 | 1 | 1.01 | 31.1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 0.0714 | 0.0485 | 0.943 | 0.811 | 0.196 | 0.973 | 0.0223 | 2.13 | 23.7 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.517 | 1.09 | 0.979 | 1 | 21.2 | 1 | 1 | 242 | 148 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.516 | 1.09 | 1 | 1 | 21.2 | 1 | 1 | 0.992 | 5.24 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 4.70e-03 | 3.41e-03 | 0.059 | 0.0529 | 0.0354 | 0.0625 | 2.65e-03 | 0.16 | 0.106 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.51 | 1.07 | 1 | 1 | 20.6 | 1 | 1 | 1.78 | 7.63e+04 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.516 | 1.09 | 1 | 1 | 21.2 | 1 | 1 | 22.4 | 2.11e+03 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 05

$T_0$ = 1049.0 K  $P_0$ = 4.666 atm  $\phi_0$ = 0.674  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=1000) · Soft long (lambda=1000)




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 05 | Cantera vs NN NRMSE

T0=1049.0 K  P0=4.666 atm  phi0=0.674  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | **base10k** | 0.299 | 0.773 | 1.07 | 1.02 | 0.614 | 11.4 | 0.509 | 1.03 | 1.01 | 0 | Complete |
| **1 ANN10** | **soft10k** | 0.752 | 1.36 | 1 | 37.2 | 0.686 | 6.97 | 1 | 1 | 2.97 | 0 | Complete |
| **1 ANN10** | **long** | 0.693 | 0.631 | 1 | 0.864 | 0.395 | 24.8 | 1 | 1 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | **base10k** | 0.565 | 1.48 | 1 | 1 | 0.929 | 1 | 1 | 0.997 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | **soft10k** | 0.58 | 1 | 1 | 1 | 0.304 | 1 | 1 | 2.83e+03 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | **long** | 0.577 | 1.1 | 1 | 1 | 0.377 | 1 | 1 | 2.46e+03 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **base10k** | 0.565 | 1.48 | 1 | 1 | 0.929 | 1 | 1 | 0.997 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **soft10k** | 0.564 | 1.47 | 1 | 0.996 | 0.926 | 1 | 1 | 0.998 | 3.21 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **long** | 0.608 | 1.23 | 1 | 0.853 | 0.705 | 11.4 | 1 | 1 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | **base10k** | 0.35 | 0.942 | 2.92 | 0.887 | 0.604 | 1.32 | 0.626 | 0.989 | 0.997 | 0 | Complete |
| **4 T/P DeepONet10** | **soft10k** | 0.565 | 1.48 | 1 | 1 | 0.928 | 1 | 1 | 1.48 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | **long** | 0.565 | 1.48 | 1 | 1 | 0.929 | 1 | 1 | 0.978 | 0.987 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **base10k** | 0.34 | 0.913 | 2.09 | 0.728 | 0.586 | 1.34 | 0.606 | 0.921 | 0.996 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **soft10k** | 0.565 | 1.47 | 1 | 0.993 | 0.927 | 1 | 1 | 0.997 | 2.21 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **long** | 0.565 | 1.48 | 1 | 0.947 | 0.928 | 1 | 1 | 1 | 5.92 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **base10k** | 0.565 | 1.48 | 1 | 1 | 0.929 | 1 | 1 | 0.999 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **soft10k** | 0.565 | 1.47 | 1 | 1 | 0.927 | 1 | 1 | 5.67 | 1.27 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **long** | 0.565 | 1.48 | 1 | 1 | 0.929 | 1 | 1 | 0.987 | 0.993 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **base10k** | 0.565 | 1.48 | 1 | 1 | 0.929 | 1 | 1 | 0.998 | 1 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **soft10k** | 0.567 | 1.47 | 1 | 1.53 | 0.923 | 1 | 1 | 0.995 | 2.73 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **long** | 0.565 | 1.48 | 1 | 1 | 0.929 | 1 | 1 | 0.999 | 1 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 07

$T_0$ = 930.6 K  $P_0$ = 3.722 atm  $\phi_0$ = 0.754  |  Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=1000) · Soft long (lambda=1000)

**1 ANN10**

**2 P-only DeepONet10**

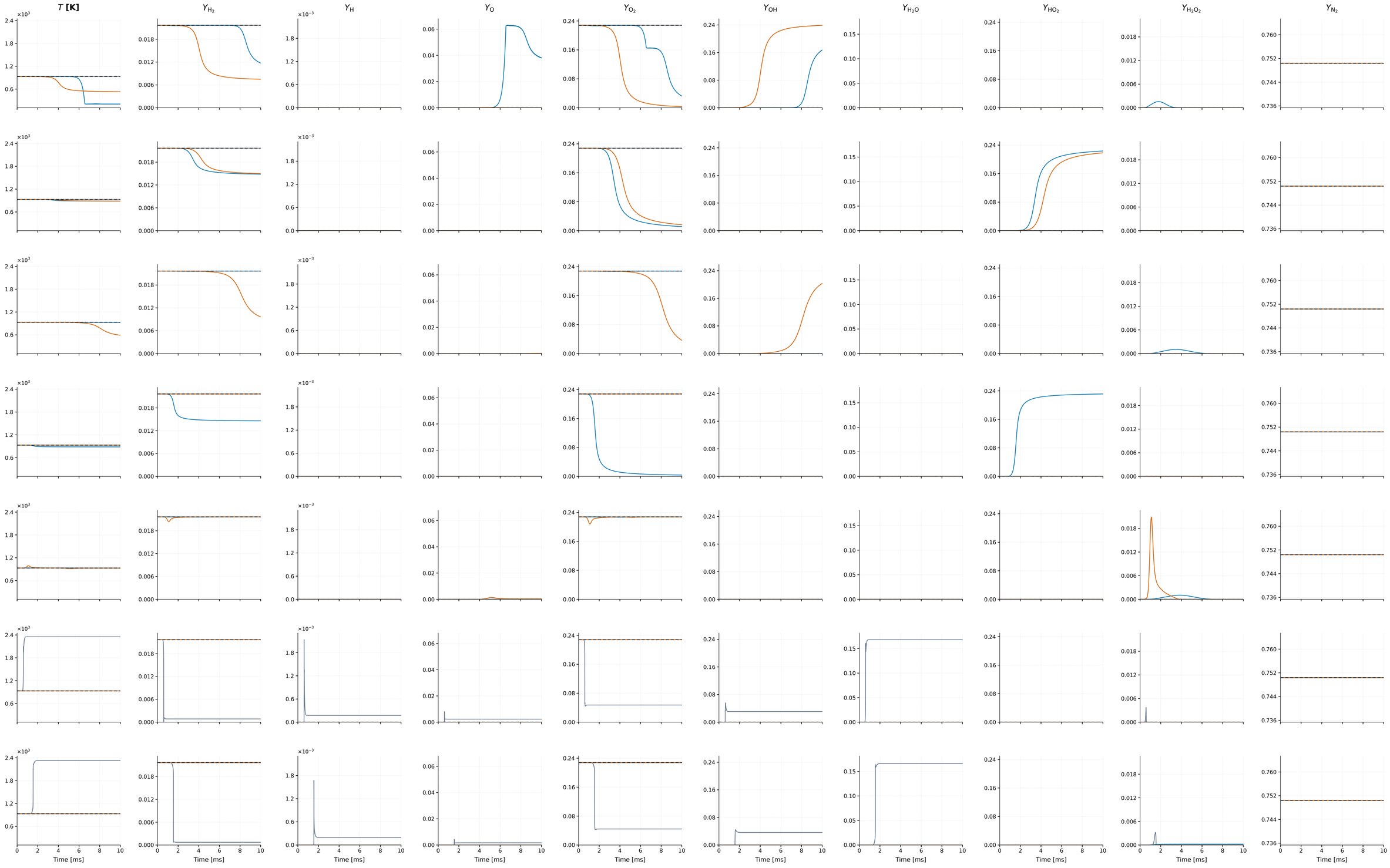
**3 ANN10 + P-only DeepONet10**

**4 T/P DeepONet10**

**5 ANN10 + T/P DeepONet10**

**6 T/P/$\phi$ DeepONet10**

**7 ANN10 + T/P/$\phi$ DeepONet10**




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 07 | Cantera vs NN NRMSE

T0=930.6 K  P0=3.722 atm  phi0=0.754  |  Repaired self-fed,10ms

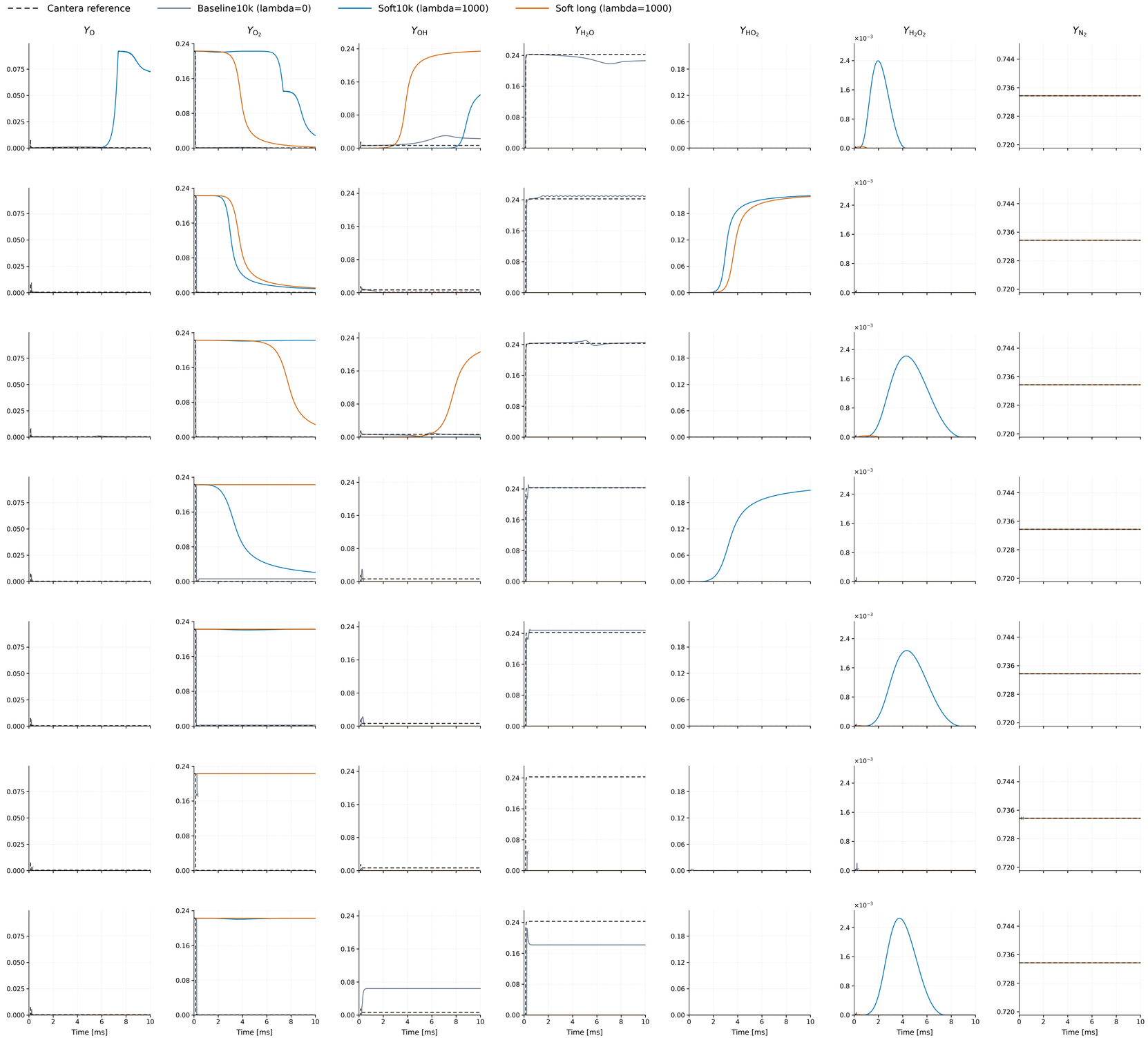
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 2.26e-06 | 1.72e-06 | 1 | 360 | 2.48e-06 | 1 | 1 | 0.531 | 0.464 | 0 | Complete |
| **1 ANN10** | soft10k | 0.463 | 0.147 | 1 | 4.38e+09 | 0.323 | 1.19e+10 | 1 | 1 | 1.35e+03 | 0 | Complete |
| **1 ANN10** | long | 0.309 | 0.471 | 1 | 6.79e+06 | 0.711 | 3.82e+10 | 1 | 1 | 14.3 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 2.84e-06 | 2.61e-06 | 1 | 0.68 | 4.31e-06 | 0.986 | 0.999 | 0.994 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.0356 | 0.235 | 1 | 1 | 0.71 | 1 | 1 | 3.45e+05 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.0319 | 0.211 | 1 | 1 | 0.636 | 1 | 1 | 3.09e+05 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 2.84e-06 | 2.61e-06 | 1 | 0.623 | 4.31e-06 | 0.987 | 0.999 | 0.992 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 1.54e-03 | 1.32e-03 | 1 | 3.98e+06 | 1.99e-03 | 2.07e+04 | 1 | 0.999 | 1.22e+03 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.137 | 0.21 | 1 | 9.26e+06 | 0.317 | 1.70e+10 | 1 | 0.998 | 17 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 2.85e-06 | 2.61e-06 | 29.5 | 0.994 | 4.31e-06 | 1.56 | 1 | 0.993 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.0435 | 0.287 | 1 | 1 | 0.867 | 1 | 1 | 4.21e+05 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 2.87e-05 | 2.99e-05 | 1 | 1 | 5.11e-05 | 1 | 1 | 5.88 | 24.4 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 2.86e-06 | 2.49e-06 | 105 | 0.972 | 4.01e-06 | 14.1 | 0.99 | 0.861 | 0.984 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 1.55e-03 | 1.36e-03 | 1 | 1.32e+06 | 2.05e-03 | 1.48e+04 | 1 | 0.996 | 1.26e+03 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.0128 | 0.0103 | 2.71e+04 | 6.14e+07 | 0.0157 | 5.98e+05 | 0.216 | 157 | 9.45e+03 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 1.48 | 0.931 | 5.41e+07 | 3.02e+08 | 0.768 | 6.67e+09 | 9.77e+05 | 107 | 599 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 6.13e-06 | 2.48e-05 | 1 | 0.974 | 7.69e-05 | 1 | 1 | 38.1 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 7.35e-06 | 7.51e-06 | 1 | 1.03 | 1.26e-05 | 1 | 1 | 1.1 | 6.83 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 1.38 | 0.888 | 5.23e+07 | 2.11e+08 | 0.74 | 7.44e+09 | 9.16e+05 | 119 | 900 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 6.38e-04 | 5.63e-04 | 1 | 1 | 8.49e-04 | 1 | 1 | 0.603 | 521 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 8.02e-05 | 5.86e-06 | 2.93e+04 | 7.15e+05 | 2.11e-05 | 27.8 | 0.985 | 1.38 | 3.36 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 18

$T_0$ = 1069.1 K  $P_0$ = 2.170 atm  $\phi_0$ = 1.544  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 18 | Cantera vs NN NRMSE

T0=1069.1 K  P0=2.170 atm  phi0=1.544  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

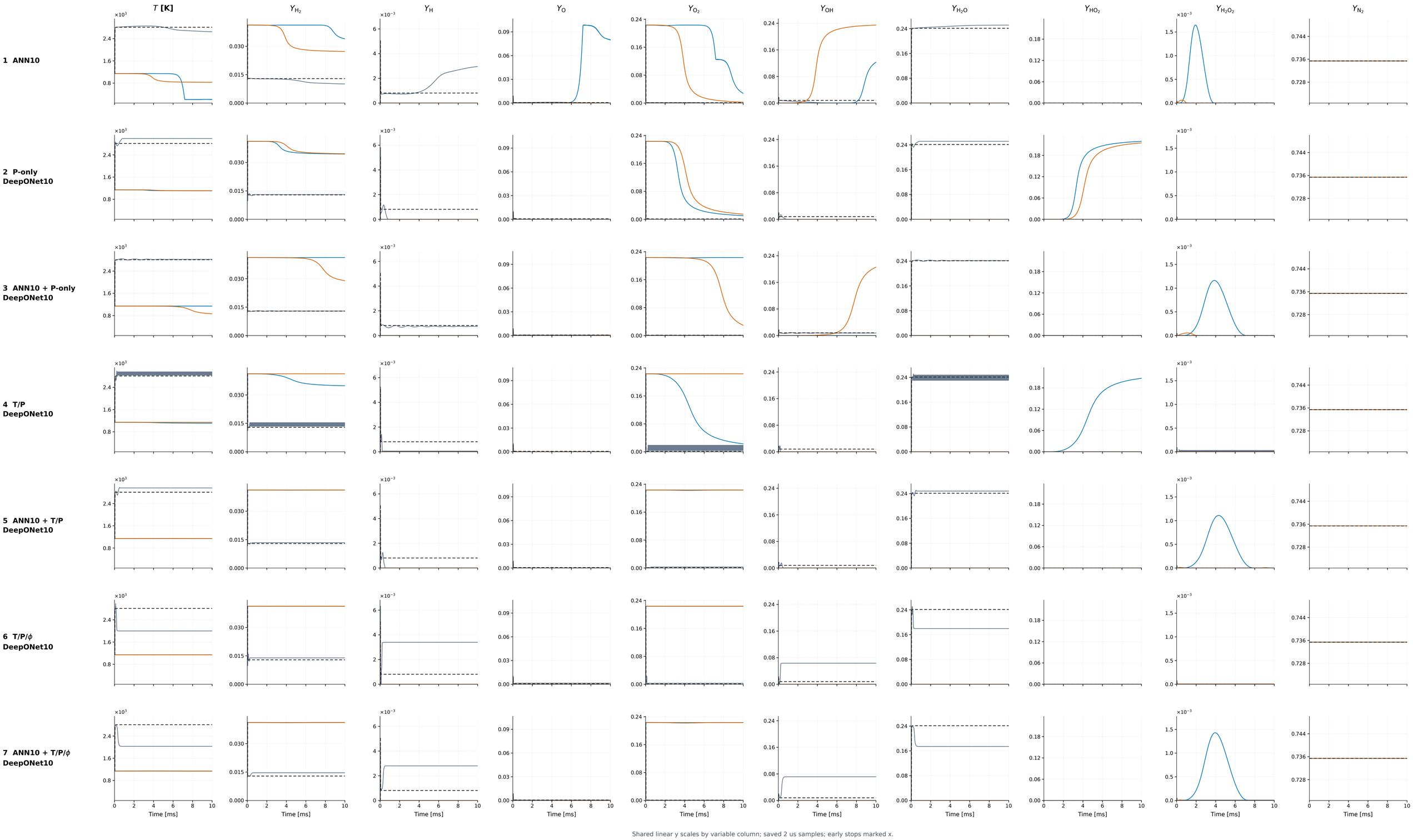
| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | **base10k** | 0.0255 | 0.115 | 0.793 | 0.959 | 0.336 | 1.94 | 0.0671 | 1.02 | 1.19 | 0 | Complete |
| **1 ANN10** | **soft10k** | 0.714 | 1.75 | 1 | 89.1 | 7.69 | 5.29 | 1 | 1 | 756 | 0 | Complete |
| **1 ANN10** | **long** | 0.677 | 1.33 | 1 | 1 | 5.27 | 25.1 | 1 | 0.999 | 9.09 | 0 | Complete |
| **2 P-only DeepONet10** | **base10k** | 0.067 | 0.161 | 0.806 | 1.05 | 0.768 | 0.82 | 0.0892 | 1.54 | 2.36 | 0 | Complete |
| **2 P-only DeepONet10** | **soft10k** | 0.616 | 1.52 | 1 | 1 | 4.73 | 1 | 1 | 1.21e+04 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | **long** | 0.615 | 1.56 | 1 | 1 | 5.24 | 1 | 1 | 1.13e+04 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **base10k** | 0.029 | 0.0992 | 0.314 | 0.756 | 0.462 | 0.297 | 0.051 | 1.31 | 1.86 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **soft10k** | 0.607 | 1.8 | 1 | 1 | 8.81 | 1 | 1 | 1 | 1.01e+03 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **long** | 0.631 | 1.64 | 1 | 0.999 | 7.56 | 12.4 | 1 | 1 | 10.6 | 0 | Complete |
| **4 T/P DeepONet10** | **base10k** | 0.0652 | 0.169 | 1.06 | 1.06 | 0.778 | 1.03 | 0.0828 | 1.98 | 5.67 | 0 | Complete |
| **4 T/P DeepONet10** | **soft10k** | 0.615 | 1.56 | 1 | 1 | 4.96 | 1 | 1 | 1.07e+04 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | **long** | 0.608 | 1.8 | 1 | 1 | 8.84 | 1 | 1 | 0.998 | 1.81 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **base10k** | 0.0681 | 0.149 | 1.03 | 1.02 | 0.693 | 1.02 | 0.0796 | 1.62 | 2.2 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **soft10k** | 0.607 | 1.8 | 1 | 1 | 8.81 | 1 | 1 | 1 | 913 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **long** | 0.608 | 1.8 | 1 | 0.999 | 8.84 | 1 | 1 | 0.999 | 2.52 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **base10k** | — | — | — | — | — | — | — | — | — | — | Stop 0.250 ms |
| **6 T/P/$\phi$ DeepONet10** | **soft10k** | 0.608 | 1.8 | 1 | 1 | 8.84 | 1 | 1 | 1.17 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **long** | 0.608 | 1.8 | 1 | 1 | 8.84 | 1 | 1 | 0.998 | 2.75 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **base10k** | 0.304 | 0.175 | 3.44 | 0.629 | 0.833 | 8.52 | 0.265 | 2.04 | 5.77 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **soft10k** | 0.607 | 1.8 | 1 | 0.991 | 8.81 | 1 | 1 | 1 | 1.06e+03 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **long** | 0.608 | 1.8 | 1 | 0.836 | 8.83 | 1 | 1 | 0.999 | 2.53 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 32

$T_0$ = 1140.7 K  $P_0$ = 3.354 atm  $\phi_0$ = 1.463  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=1000) · Soft long (lambda=1000)




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 32 | Cantera vs NN NRMSE

T0=1140.7 K  P0=3.354 atm  phi0=1.463  |  Repaired self-fed,10ms

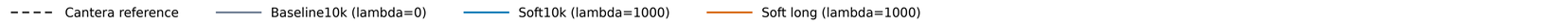
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

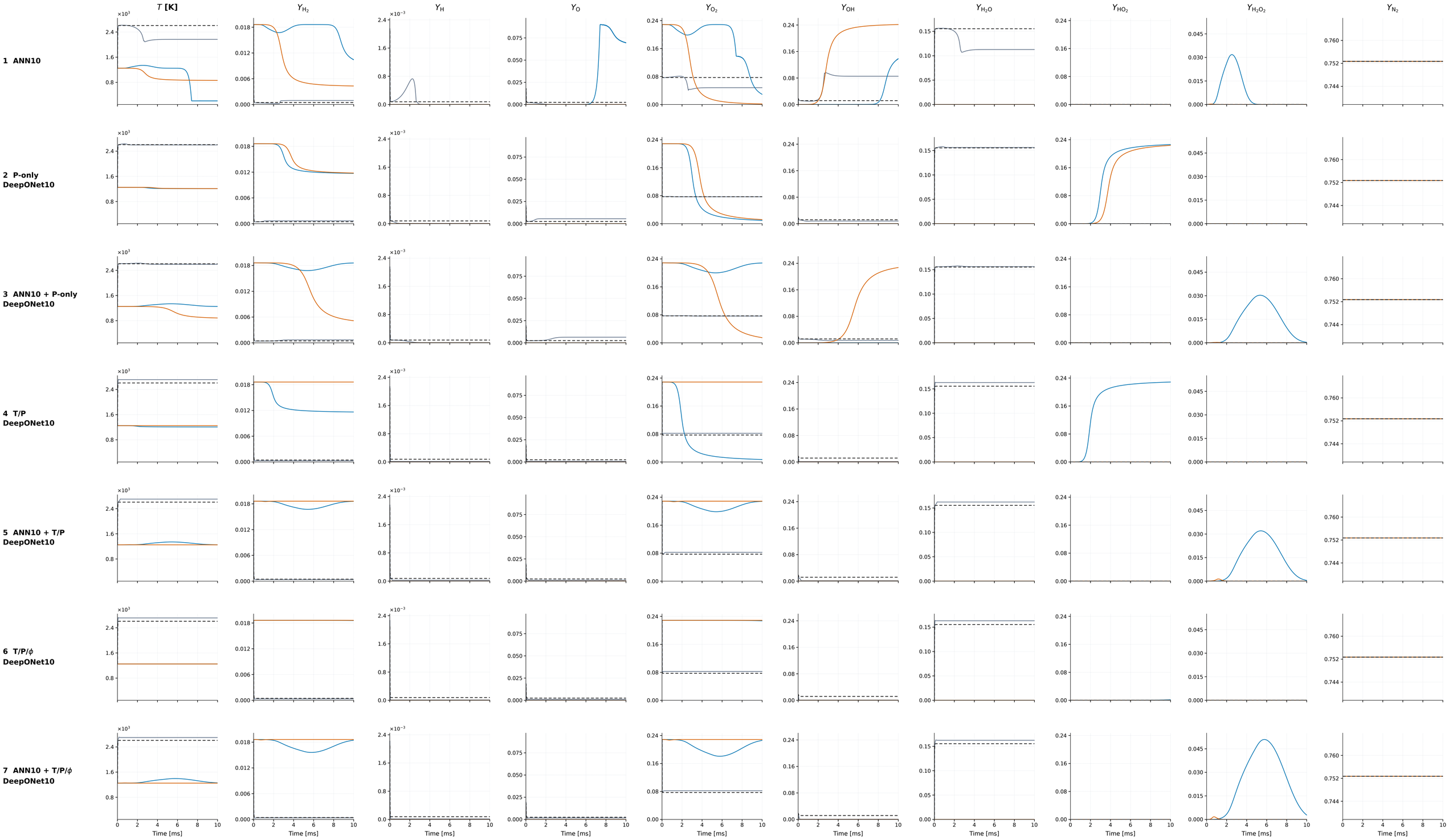
| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | **base10k** | 0.0312 | 0.126 | 1.48 | 0.511 | 0.0533 | 0.83 | 0.0317 | 0.118 | 1.03 | 0 | Complete |
| **1 ANN10** | **soft10k** | 0.712 | 2.09 | 1 | 78.4 | 13.9 | 4.19 | 1 | 0.999 | 653 | 0 | Complete |
| **1 ANN10** | **long** | 0.661 | 1.6 | 1 | 0.857 | 9.75 | 20.4 | 1 | 1 | 16.2 | 0 | Complete |
| **2 P-only DeepONet10** | **base10k** | 0.0629 | 0.063 | 0.97 | 0.99 | 0.447 | 0.972 | 0.0492 | 1.24 | 1.51 | 0 | Complete |
| **2 P-only DeepONet10** | **soft10k** | 0.601 | 1.84 | 1 | 1 | 9.17 | 1 | 1 | 1.49e+04 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | **long** | 0.6 | 1.89 | 1 | 1 | 10.2 | 1 | 1 | 1.35e+04 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **base10k** | 8.62e-03 | 0.0401 | 0.15 | 0.275 | 0.276 | 0.0993 | 0.0162 | 0.778 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **soft10k** | 0.593 | 2.15 | 1 | 0.997 | 16.1 | 1 | 1 | 0.999 | 608 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **long** | 0.617 | 1.97 | 1 | 0.938 | 13.9 | 10 | 1 | 1 | 19.5 | 0 | Complete |
| **4 T/P DeepONet10** | **base10k** | 0.04 | 0.127 | 0.968 | 1 | 0.844 | 0.993 | 0.0346 | 2.01 | 13 | 0 | Complete |
| **4 T/P DeepONet10** | **soft10k** | 0.6 | 1.91 | 1 | 1 | 10.3 | 1 | 1 | 1.23e+04 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | **long** | 0.594 | 2.15 | 1 | 1 | 16.2 | 1 | 1 | 0.994 | 1.98 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **base10k** | 0.0538 | 0.0417 | 0.965 | 0.912 | 0.184 | 0.98 | 0.0292 | 0.891 | 0.997 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **soft10k** | 0.593 | 2.15 | 1 | 1 | 16.1 | 1 | 1 | 0.999 | 609 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **long** | 0.594 | 2.15 | 1 | 1 | 16.2 | 1 | 1 | 0.997 | 3.25 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **base10k** | 0.285 | 0.0903 | 3.13 | 1.31 | 0.345 | 6.75 | 0.254 | 2.61 | 7.38 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **soft10k** | 0.594 | 2.15 | 1 | 1 | 16.2 | 1 | 1 | 1.47 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **long** | 0.594 | 2.15 | 1 | 1 | 16.2 | 1 | 1 | 0.994 | 3.6 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **base10k** | 0.272 | 0.13 | 2.39 | 0.126 | 0.0193 | 7.65 | 0.272 | 1.92 | 9.92 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **soft10k** | 0.593 | 2.15 | 1 | 0.998 | 16.1 | 1 | 1 | 0.999 | 752 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **long** | 0.594 | 2.15 | 1 | 0.851 | 16.2 | 1 | 1 | 0.999 | 2.82 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 39

$T_0$ = 1250.1 K  $P_0$ = 2.297 atm  $\phi_0$ = 0.646  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 39 | Cantera vs NN NRMSE

T0=1250.1 K  P0=2.297 atm  phi0=0.646  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 0.15 | 0.476 | 2.07 | 0.897 | 0.329 | 5.49 | 0.241 | 0.78 | 0.837 | 0 | Complete |
| **1 ANN10** | soft10k | 0.651 | 19.1 | 1 | 16.3 | 1.59 | 3.32 | 1 | 0.991 | 1.26e+04 | 0 | Complete |
| **1 ANN10** | long | 0.626 | 11.4 | 1 | 1 | 1.18 | 15.5 | 1 | 0.998 | 59.6 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 3.93e-03 | 0.236 | 0.758 | 1.17 | 6.65e-03 | 0.321 | 6.61e-03 | 0.331 | 0.24 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.531 | 15.7 | 1 | 1 | 1.18 | 1 | 1 | 9.64e+03 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.529 | 16.3 | 1 | 1 | 1.25 | 1 | 1 | 8.88e+03 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 5.60e-03 | 0.233 | 0.703 | 1.3 | 7.30e-03 | 0.33 | 6.99e-03 | 0.292 | 0.178 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.506 | 19.7 | 1 | 1 | 1.8 | 0.999 | 1 | 0.979 | 1.72e+04 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.581 | 15.6 | 1 | 1 | 1.4 | 10.6 | 1 | 0.996 | 73.3 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 0.0406 | 0.212 | 0.694 | 0.675 | 0.0647 | 0.897 | 0.0488 | 0.769 | 0.794 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.532 | 14.7 | 1 | 1 | 1.05 | 1 | 1 | 1.05e+04 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.52 | 20.5 | 1 | 1 | 1.95 | 1 | 1 | 0.861 | 4.95 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 0.0362 | 0.13 | 0.568 | 0.562 | 0.0688 | 0.891 | 0.0437 | 0.833 | 0.881 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.505 | 19.6 | 1 | 1 | 1.79 | 0.999 | 1 | 0.976 | 1.82e+04 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.52 | 20.5 | 1 | 0.992 | 1.95 | 1 | 1 | 1.57 | 299 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 0.0408 | 0.197 | 0.787 | 0.562 | 0.0679 | 0.955 | 0.0499 | 0.634 | 0.567 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.52 | 20.5 | 1 | 1 | 1.95 | 1 | 1 | 18.9 | 7.53 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.52 | 20.5 | 1 | 1 | 1.95 | 1 | 1 | 0.954 | 4.92 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 0.0351 | 0.121 | 0.697 | 0.309 | 0.0652 | 0.934 | 0.0446 | 0.686 | 0.692 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.495 | 19 | 1 | 1 | 1.68 | 1 | 1 | 0.949 | 3.01e+04 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.52 | 20.5 | 1 | 0.988 | 1.95 | 1 | 1 | 1.71 | 336 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 44

$T_0$ = 1031.1 K  $P_0$ = 2.272 atm  $\phi_0$ = 0.770  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference  Baseline10k (lambda=0)  Soft10k (lambda=1000)  Soft long (lambda=1000)

**1 ANN10**

**2 P-only DeepONet10**

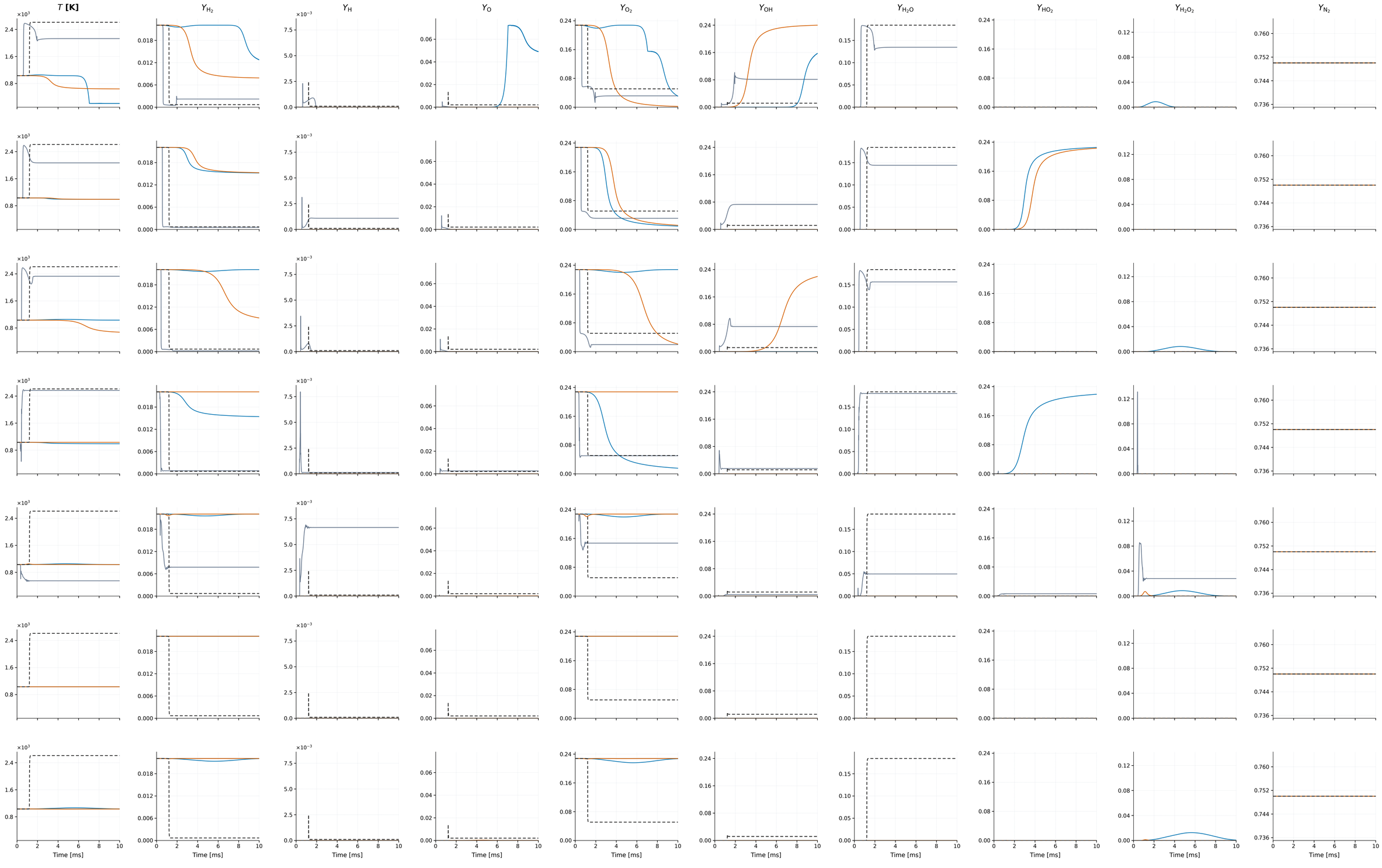
**3 ANN10 + P-only DeepONet10**

**4 T/P DeepONet10**

**5 ANN10 + T/P DeepONet10**

**6 T/P/$\phi$ DeepONet10**

**7 ANN10 + T/P/$\phi$ DeepONet10**




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 44 | Cantera vs NN NRMSE

T0=1031.1 K  P0=2.272 atm  phi0=0.770  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

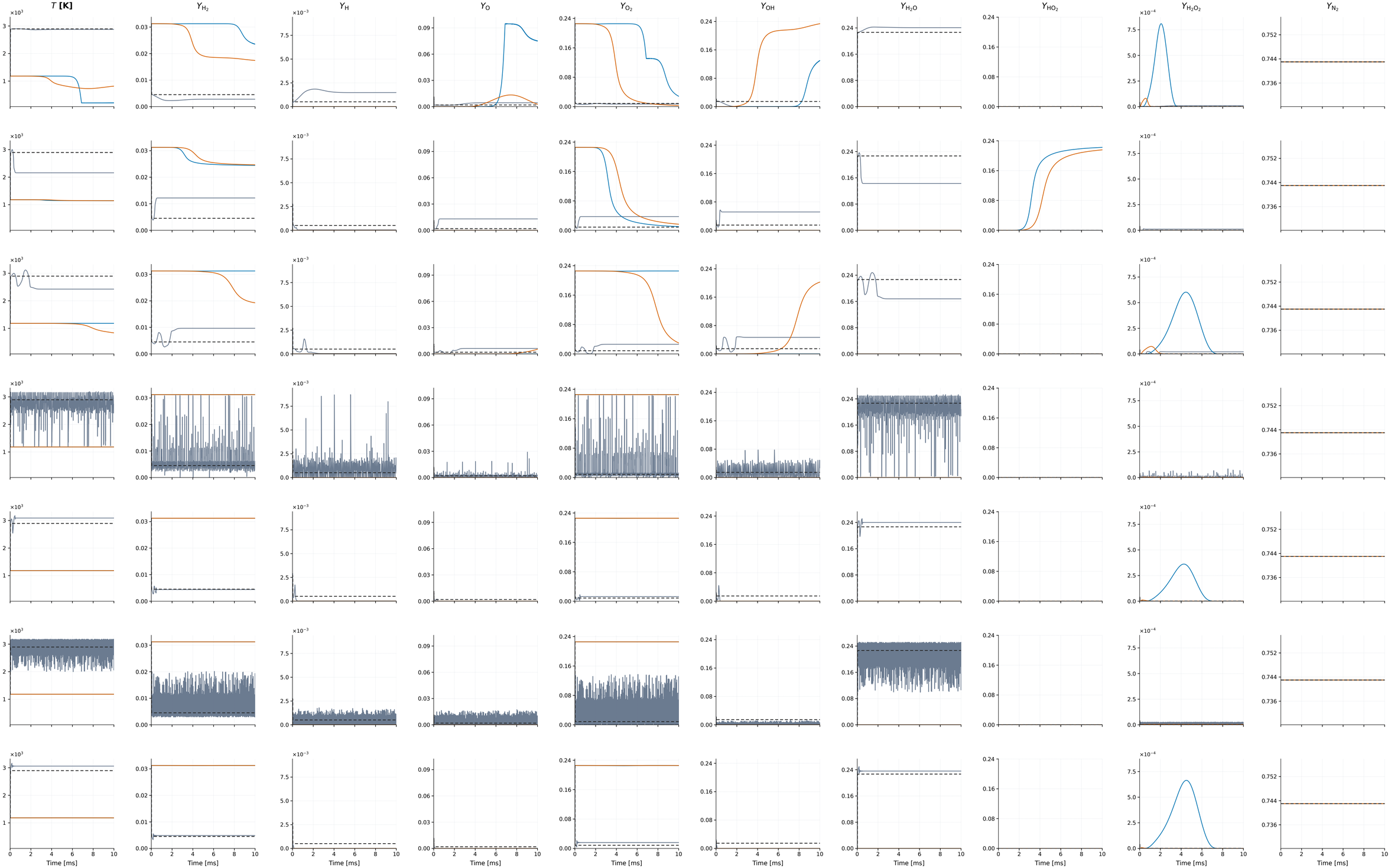
| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 0.231 | 0.688 | 2.26 | 0.999 | 0.481 | 5.74 | 0.363 | 1.11 | 1.23 | 0 | Complete |
| **1 ANN10** | soft10k | 0.725 | 2.45 | 1 | 17 | 1.47 | 4.14 | 1 | 0.999 | 398 | 0 | Complete |
| **1 ANN10** | long | 0.708 | 1.49 | 1 | 1 | 0.86 | 15.8 | 1 | 1 | 1.93 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 0.252 | 0.701 | 8.52 | 0.997 | 0.528 | 5.19 | 0.335 | 1.82 | 1.38 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.611 | 1.99 | 1 | 1 | 0.806 | 1 | 1 | 6.37e+03 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.609 | 2.07 | 1 | 1 | 0.934 | 1 | 1 | 5.87e+03 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 0.194 | 0.773 | 1.77 | 1.02 | 0.628 | 5.32 | 0.317 | 1.03 | 0.986 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.595 | 2.58 | 1 | 1 | 1.76 | 1 | 1 | 0.997 | 529 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.65 | 2.11 | 1 | 1 | 1.31 | 9.46 | 1 | 0.999 | 1.75 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 0.169 | 0.784 | 4.31 | 0.488 | 0.547 | 0.67 | 0.288 | 10.8 | 819 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.61 | 2.01 | 1 | 1 | 0.713 | 1 | 1 | 6.08e+03 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.598 | 2.6 | 1 | 1 | 1.79 | 1 | 1 | 0.925 | 1.28 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 0.782 | 0.972 | 57.9 | 0.965 | 1 | 0.638 | 0.733 | 221 | 3.74e+03 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.595 | 2.57 | 1 | 1 | 1.76 | 1 | 1 | 0.996 | 557 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.599 | 2.6 | 1 | 0.979 | 1.79 | 1 | 1 | 1.37 | 164 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 0.599 | 2.6 | 1 | 1 | 1.79 | 1 | 1 | 1 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.599 | 2.6 | 1 | 1 | 1.79 | 1 | 1 | 0.813 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.599 | 2.6 | 1 | 1 | 1.79 | 1 | 1 | 0.977 | 1.02 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 0.599 | 2.6 | 1 | 1 | 1.79 | 1 | 1 | 0.999 | 1 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.591 | 2.55 | 1 | 1 | 1.73 | 1 | 1 | 0.994 | 908 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.599 | 2.6 | 1 | 0.968 | 1.79 | 1 | 1 | 0.88 | 27.8 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 45

$T_0$ = 1181.9 K  $P_0$ = 4.842 atm  $\phi_0$ = 1.099  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference  Baseline10k (lambda=0)  Soft10k (lambda=1000)  Soft long (lambda=1000)




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 45 | Cantera vs NN NRMSE

T0=1181.9 K  P0=4.842 atm  phi0=1.099  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | **base10k** | 8.22e-03 | 0.379 | 1.98 | 1.07 | 0.214 | 0.942 | 0.0629 | 1.61 | 5.76 | 0 | Complete |
| **1 ANN10** | **soft10k** | 0.722 | 5.45 | 1 | 25.3 | 13.5 | 2.47 | 1 | 0.996 | 188 | 0 | Complete |
| **1 ANN10** | **long** | 0.679 | 4.15 | 1 | 3.02 | 9.59 | 10.6 | 1 | 0.999 | 12.1 | 0 | Complete |
| **2 P-only DeepONet10** | **base10k** | 0.25 | 1.59 | 0.904 | 5.59 | 2.08 | 2.48 | 0.364 | 2 | 7.4 | 0 | Complete |
| **2 P-only DeepONet10** | **soft10k** | 0.6 | 4.75 | 1 | 1 | 8.85 | 1 | 1 | 1.21e+04 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | **long** | 0.598 | 4.93 | 1 | 1 | 10.2 | 1 | 1 | 1.05e+04 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **base10k** | 0.148 | 0.982 | 0.896 | 1.99 | 1.19 | 2.01 | 0.236 | 2.46 | 12.9 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **soft10k** | 0.591 | 5.63 | 1 | 0.999 | 15.9 | 1 | 1 | 0.992 | 200 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **long** | 0.617 | 5.13 | 1 | 0.946 | 13.6 | 5.25 | 1 | 0.998 | 15.1 | 0 | Complete |
| **4 T/P DeepONet10** | **base10k** | 0.137 | 1.06 | 1.85 | 0.835 | 2.96 | 1.05 | 0.199 | 4.01 | 5.37 | 0 | Complete |
| **4 T/P DeepONet10** | **soft10k** | 0.592 | 5.63 | 1 | 1 | 16 | 1 | 1 | 0.803 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | **long** | 0.592 | 5.63 | 1 | 1 | 16 | 1 | 1 | 0.941 | 1.33 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **base10k** | 0.0687 | 0.168 | 0.958 | 0.974 | 0.537 | 0.996 | 0.0649 | 1.19 | 0.682 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **soft10k** | 0.591 | 5.63 | 1 | 0.999 | 15.9 | 1 | 1 | 0.988 | 118 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **long** | 0.592 | 5.63 | 1 | 1 | 16 | 1 | 1 | 0.969 | 1.56 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **base10k** | 0.0835 | 0.76 | 0.851 | 1.55 | 2.22 | 0.889 | 0.134 | 2.86 | 5.55 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **soft10k** | 0.592 | 5.63 | 1 | 1 | 16 | 1 | 1 | 0.897 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **long** | 0.592 | 5.63 | 1 | 1 | 16 | 1 | 1 | 0.946 | 1.74 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **base10k** | 0.0594 | 0.181 | 0.981 | 0.969 | 0.723 | 0.997 | 0.0499 | 0.892 | 0.616 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **soft10k** | 0.591 | 5.63 | 1 | 0.996 | 15.9 | 1 | 1 | 0.978 | 219 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **long** | 0.592 | 5.63 | 1 | 1 | 16 | 1 | 1 | 0.985 | 1 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 53

$T_0$ = 1257.8 K  $P_0$ = 2.103 atm  $\phi_0$ = 1.799  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference  Baseline10k (lambda=0)  Soft10k (lambda=1000)  Soft long (lambda=1000)

**1 ANN10**

**2 P-only DeepONet10**

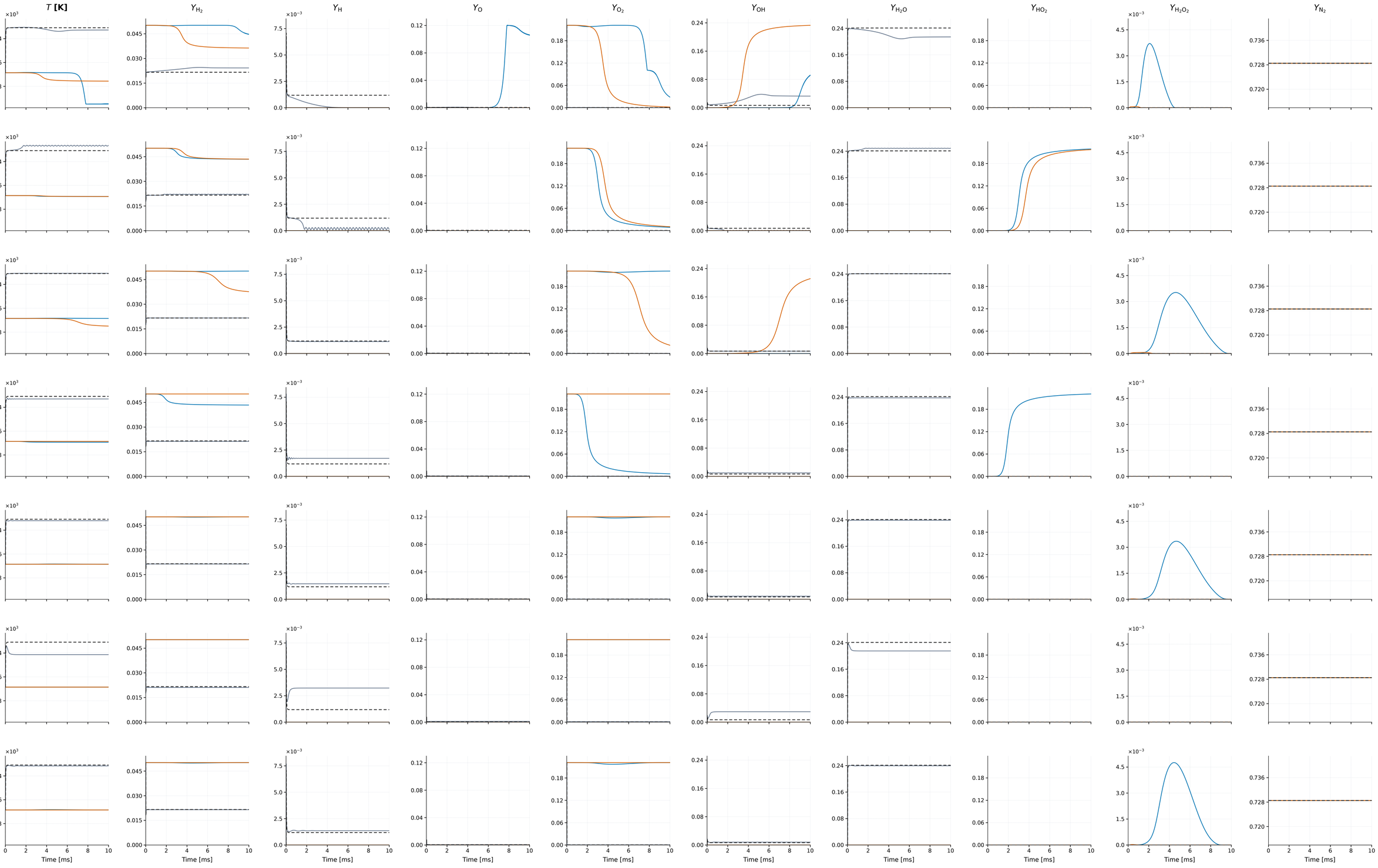
**3 ANN10 + P-only DeepONet10**

**4 T/P DeepONet10**

**5 ANN10 + T/P DeepONet10**

**6 T/P/$\phi$ DeepONet10**

**7 ANN10 + T/P/$\phi$ DeepONet10**




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 53 | Cantera vs NN NRMSE

T0=1257.8 K  P0=2.103 atm  phi0=1.799  |  Repaired self-fed,10ms

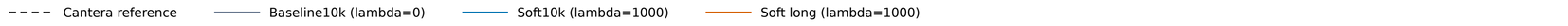
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

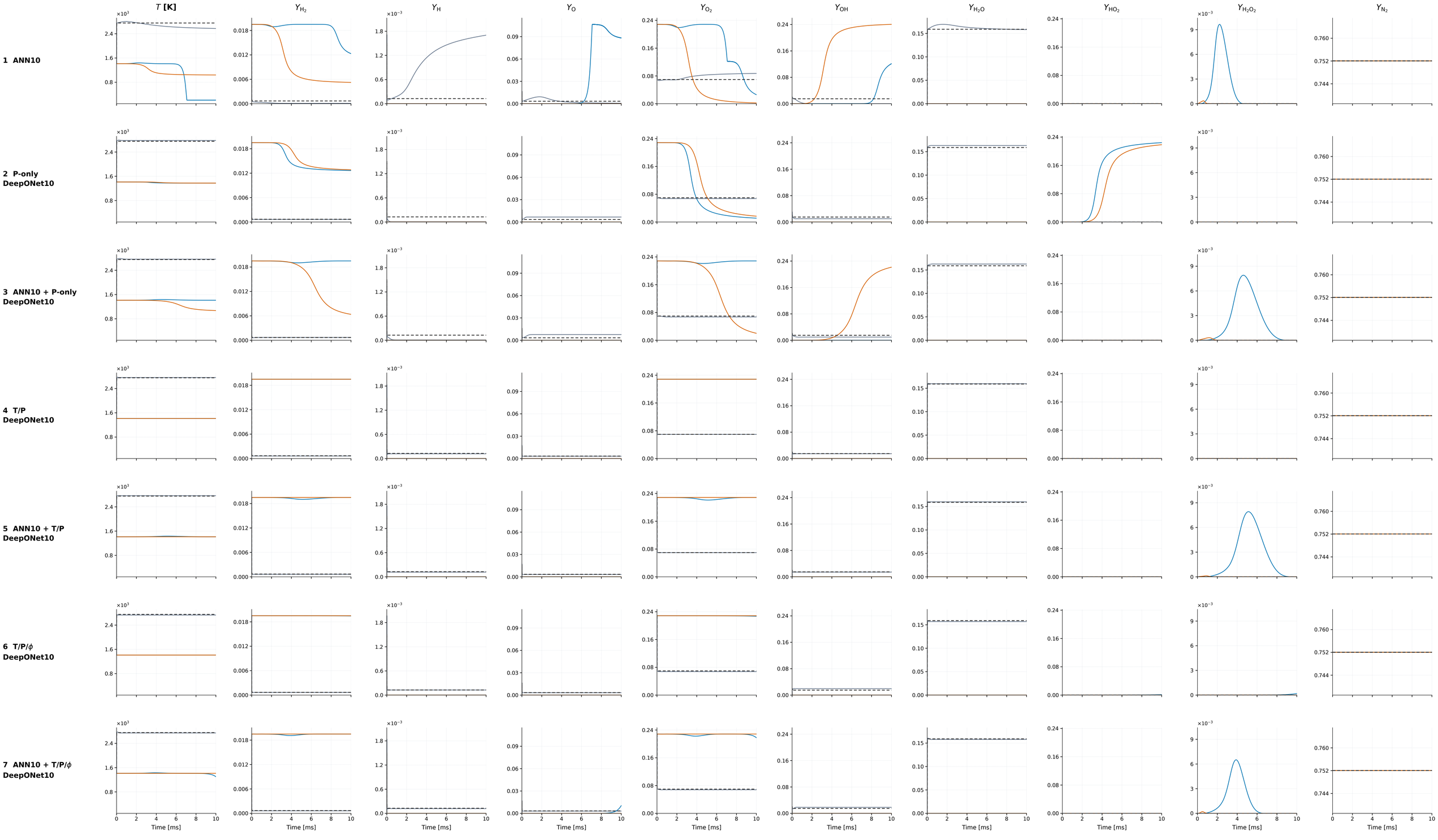
| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | **base10k** | 0.0249 | 0.105 | 0.84 | 0.642 | 0.0369 | 3.25 | 0.0957 | 0.118 | 0.566 | 0 | Complete |
| **1 ANN10** | **soft10k** | 0.66 | 1.29 | 1 | 113 | 20.4 | 3.29 | 1 | 0.999 | 6.95e+03 | 0 | Complete |
| **1 ANN10** | **long** | 0.61 | 0.959 | 1 | 1 | 13.4 | 25 | 1 | 0.999 | 95.4 | 0 | Complete |
| **2 P-only DeepONet10** | **base10k** | 0.0551 | 0.0218 | 0.754 | 0.845 | 0.0818 | 0.868 | 0.03 | 0.247 | 0.641 | 0 | Complete |
| **2 P-only DeepONet10** | **soft10k** | 0.552 | 1.12 | 1 | 1 | 12.7 | 1 | 1 | 3.38e+04 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | **long** | 0.551 | 1.14 | 1 | 1 | 13.9 | 1 | 1 | 3.16e+04 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **base10k** | 2.57e-03 | 2.74e-03 | 0.0432 | 0.0484 | 0.0322 | 0.0285 | 1.59e-03 | 0.131 | 0.184 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **soft10k** | 0.544 | 1.31 | 1 | 1 | 23.2 | 1 | 1 | 0.998 | 9.57e+03 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **long** | 0.572 | 1.17 | 1 | 0.999 | 19 | 14.2 | 1 | 0.999 | 115 | 0 | Complete |
| **4 T/P DeepONet10** | **base10k** | 0.0311 | 0.0186 | 0.439 | 0.467 | 0.196 | 0.463 | 0.0176 | 0.488 | 0.557 | 0 | Complete |
| **4 T/P DeepONet10** | **soft10k** | 0.553 | 1.08 | 1 | 1 | 9.98 | 1 | 1 | 3.70e+04 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | **long** | 0.545 | 1.32 | 1 | 1 | 23.3 | 1 | 1 | 0.994 | 3.57 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **base10k** | 0.0172 | 0.0109 | 0.236 | 0.198 | 0.13 | 0.287 | 0.0107 | 0.341 | 0.331 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **soft10k** | 0.544 | 1.31 | 1 | 1 | 23.2 | 1 | 1 | 0.998 | 8.78e+03 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **long** | 0.545 | 1.32 | 1 | 1 | 23.3 | 1 | 1 | 0.999 | 21.9 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **base10k** | 0.149 | 0.0297 | 1.67 | 1.47 | 0.293 | 3.28 | 0.106 | 1.41 | 6.57 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **soft10k** | 0.545 | 1.32 | 1 | 1 | 23.3 | 1 | 1 | 6.75 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **long** | 0.545 | 1.32 | 1 | 1 | 23.3 | 1 | 1 | 0.994 | 8.59 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **base10k** | 0.0109 | 0.0127 | 0.15 | 0.173 | 0.181 | 0.196 | 9.52e-03 | 0.429 | 0.34 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **soft10k** | 0.543 | 1.31 | 1 | 1 | 23.2 | 1 | 1 | 0.998 | 1.19e+04 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **long** | 0.545 | 1.32 | 1 | 0.994 | 23.3 | 1 | 1 | 1.01 | 26.4 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 57

$T_0$ = 1412.0 K  $P_0$ = 4.083 atm  $\phi_0$ = 0.677  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 57 | Cantera vs NN NRMSE

T0=1412.0 K  P0=4.083 atm  phi0=0.677  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 0.0428 | 0.666 | 8.62 | 0.853 | 0.176 | 0.956 | 0.0319 | 1.51 | 4.49 | 0 | Complete |
| **1 ANN10** | soft10k | 0.655 | 21.7 | 1 | 15.9 | 1.87 | 2.3 | 1 | 0.985 | 1.93e+03 | 0 | Complete |
| **1 ANN10** | long | 0.578 | 13.5 | 1 | 1 | 1.34 | 11.5 | 1 | 0.998 | 40.4 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 0.0101 | 0.104 | 0.933 | 0.986 | 0.0348 | 0.33 | 0.0267 | 0.132 | 1.47 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.496 | 18 | 1 | 1 | 1.36 | 1 | 1 | 6.24e+03 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.495 | 18.8 | 1 | 1 | 1.46 | 1 | 1 | 5.53e+03 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 5.19e-03 | 0.0455 | 0.906 | 1.27 | 0.0366 | 0.352 | 0.0232 | 0.0753 | 1.05 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.485 | 22.7 | 1 | 1 | 2.24 | 1 | 1 | 0.962 | 2.12e+03 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.533 | 18.3 | 1 | 1 | 1.69 | 7.16 | 1 | 0.996 | 50.6 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 3.80e-03 | 0.0682 | 0.0852 | 0.0645 | 2.64e-03 | 0.0271 | 4.83e-03 | 0.0327 | 0.023 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.487 | 22.8 | 1 | 1 | 2.27 | 1 | 1 | 0.561 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.487 | 22.8 | 1 | 1 | 2.27 | 1 | 1 | 0.9 | 1.84 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 4.76e-03 | 0.0786 | 0.0856 | 0.0623 | 1.90e-03 | 0.045 | 6.40e-03 | 0.0294 | 0.0218 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.485 | 22.6 | 1 | 1 | 2.24 | 1 | 1 | 0.927 | 2.15e+03 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.487 | 22.8 | 1 | 1 | 2.27 | 1 | 1 | 0.882 | 16.9 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 8.28e-03 | 0.0376 | 0.0669 | 0.036 | 0.03 | 0.288 | 0.0133 | 0.265 | 0.371 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.487 | 22.8 | 0.999 | 1 | 2.27 | 1 | 1 | 12.1 | 24.7 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.487 | 22.8 | 1 | 1 | 2.27 | 1 | 1 | 0.947 | 1.88 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 3.84e-03 | 0.0728 | 0.087 | 0.0721 | 0.0256 | 0.213 | 7.18e-03 | 0.213 | 0.325 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.488 | 22.7 | 1 | 0.973 | 2.24 | 1 | 1 | 0.93 | 1.44e+03 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.487 | 22.8 | 1 | 1 | 2.27 | 1 | 1 | 1.01 | 21.7 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 64

$T_0$ = 1045.3 K  $P_0$ = 4.301 atm  $\phi_0$ = 0.595  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference  Baseline10k (lambda=0)  Soft10k (lambda=1000)  Soft long (lambda=1000)

**1 ANN10**

**2 P-only DeepONet10**

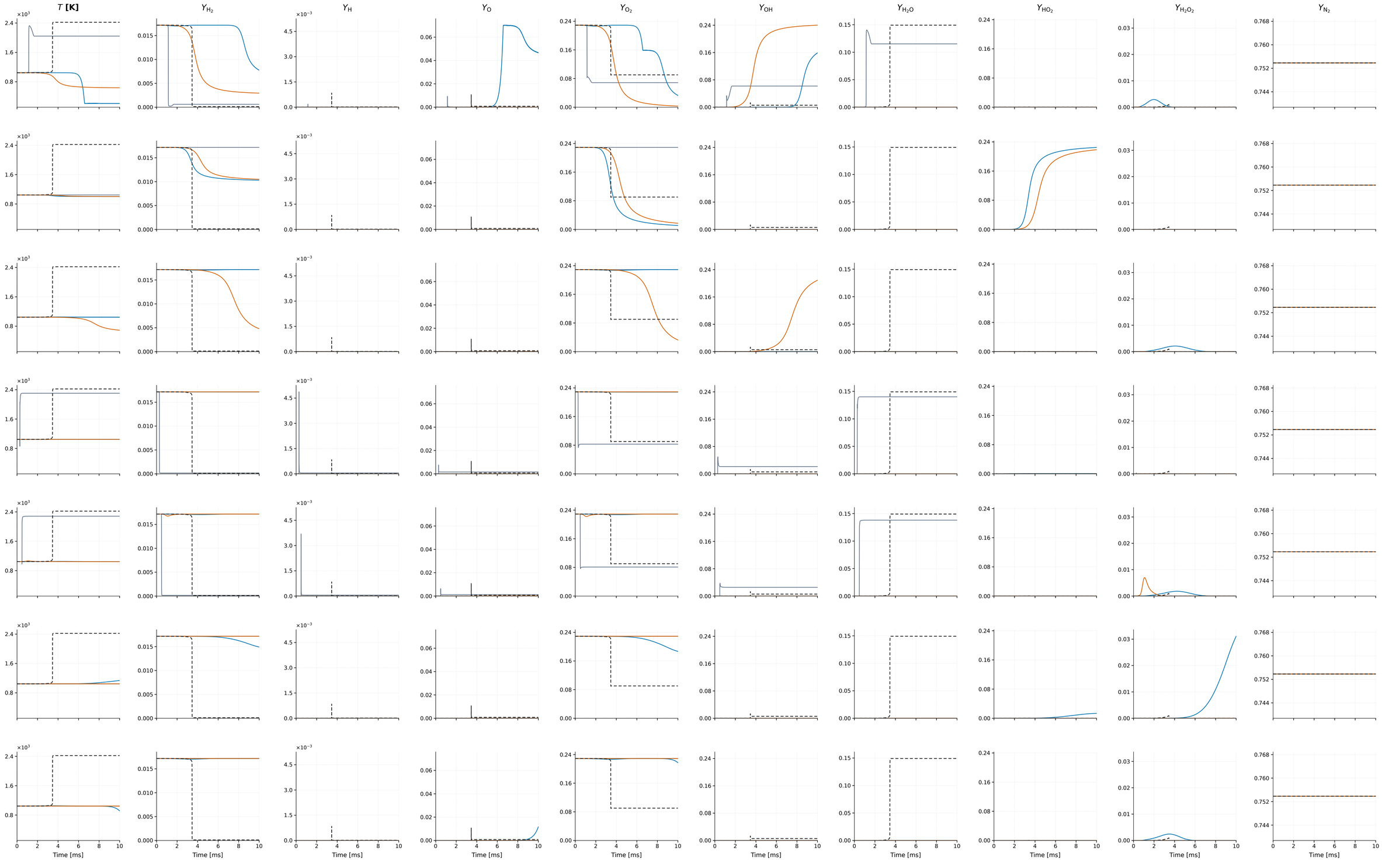
**3 ANN10 + P-only DeepONet10**

**4 T/P DeepONet10**

**5 ANN10 + T/P DeepONet10**

**6 T/P/$\phi$ DeepONet10**

**7 ANN10 + T/P/$\phi$ DeepONet10**




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 64 | Cantera vs NN NRMSE

T0=1045.3 K  P0=4.301 atm  phi0=0.595  |  Repaired self-fed,10ms

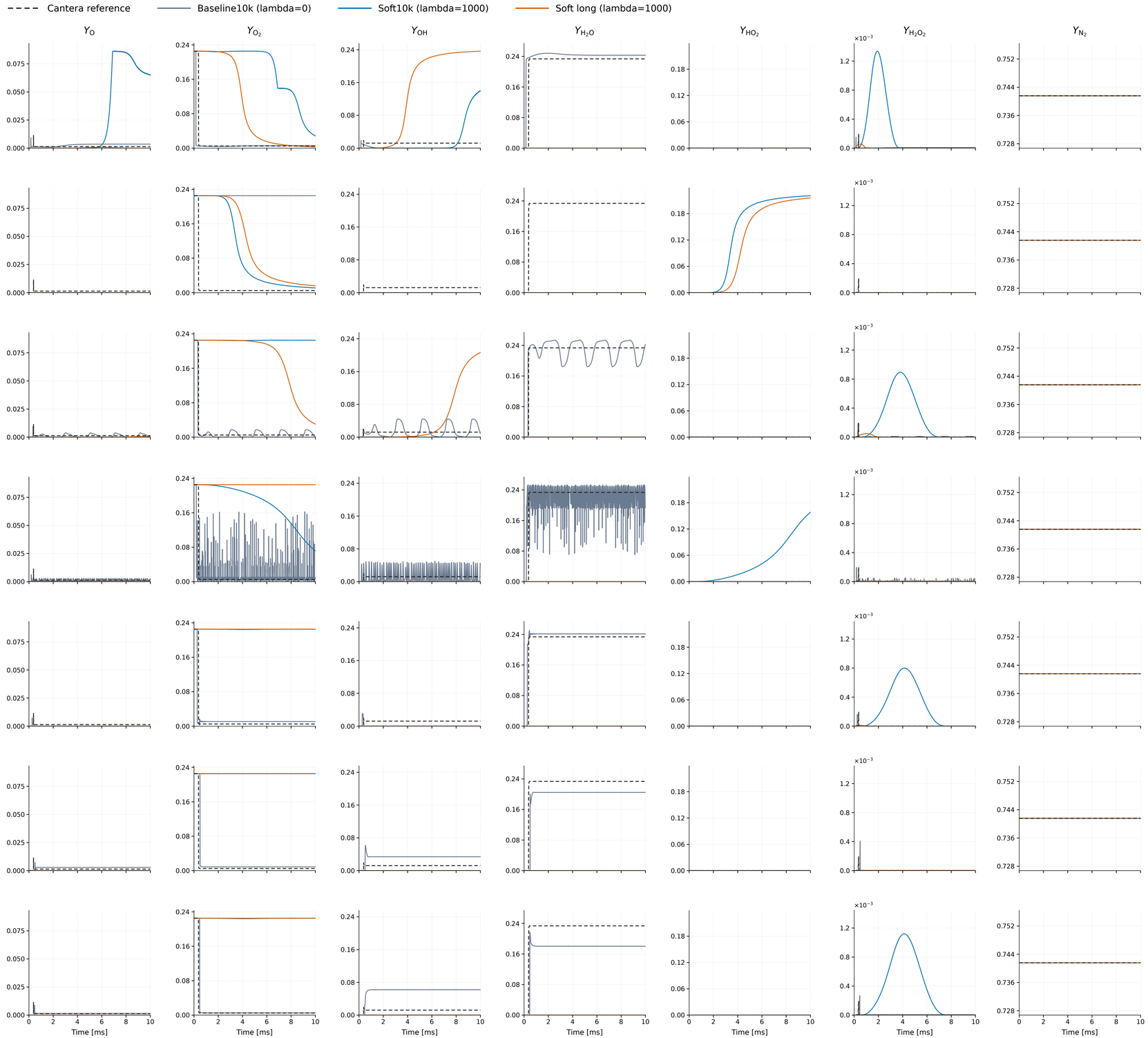
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 0.284 | 0.791 | 1.02 | 1.04 | 0.509 | 11.4 | 0.523 | 1.02 | 1.01 | 0 | Complete |
| **1 ANN10** | soft10k | 0.743 | 1.24 | 1 | 54.8 | 0.529 | 10.5 | 1 | 1 | 6.4 | 0 | Complete |
| **1 ANN10** | long | 0.686 | 0.4 | 1 | 0.987 | 0.389 | 36.9 | 1 | 1 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 0.542 | 1.37 | 1 | 1 | 0.733 | 1 | 1 | 0.999 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.558 | 0.865 | 1 | 1 | 0.361 | 1 | 1 | 2.9e+03 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.555 | 0.952 | 1 | 1 | 0.332 | 1 | 1 | 2.55e+03 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 0.542 | 1.37 | 1 | 1 | 0.733 | 1 | 1 | 0.999 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.541 | 1.37 | 1 | 0.993 | 0.73 | 1 | 1 | 0.998 | 6.28 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.597 | 1.06 | 1 | 0.97 | 0.513 | 18.6 | 1 | 1 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 0.346 | 0.949 | 6.56 | 1.71 | 0.539 | 3.79 | 0.654 | 1.21 | 0.992 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.542 | 1.37 | 1 | 1 | 0.73 | 1 | 1 | 9.66 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.542 | 1.37 | 1 | 1 | 0.733 | 1 | 1 | 0.977 | 0.984 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 0.331 | 0.919 | 4.92 | 1.11 | 0.528 | 4.59 | 0.625 | 1.09 | 0.99 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.541 | 1.37 | 1 | 0.995 | 0.73 | 1 | 1 | 0.997 | 5.39 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.542 | 1.37 | 1 | 0.883 | 0.733 | 1 | 1 | 1.16 | 10.2 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 0.542 | 1.37 | 1 | 1 | 0.733 | 1 | 1 | 0.999 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.533 | 1.32 | 0.994 | 1 | 0.668 | 1 | 1 | 101 | 67.9 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.542 | 1.37 | 1 | 1 | 0.733 | 1 | 1 | 0.987 | 0.991 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 0.542 | 1.37 | 1 | 1 | 0.733 | 1 | 1 | 0.998 | 1 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.545 | 1.37 | 1 | 2.51 | 0.727 | 1 | 1 | 0.996 | 6.26 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.542 | 1.37 | 1 | 0.968 | 0.733 | 1 | 1 | 1 | 1 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 68

$T_0$ = 1086.3 K  $P_0$ = 3.702 atm  $\phi_0$ = 1.168  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 68 | Cantera vs NN NRMSE

T0=1086.3 K  P0=3.702 atm  phi0=1.168  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 0.092 | 0.499 | 1.5 | 1.33 | 0.738 | 0.979 | 0.157 | 1.47 | 1.33 | 0 | Complete |
| **1 ANN10** | soft10k | 0.737 | 3.04 | 1 | 32.2 | 4.19 | 3.35 | 1 | 0.999 | 52.4 | 0 | Complete |
| **1 ANN10** | long | 0.691 | 2.29 | 1 | 0.878 | 2.89 | 13.6 | 1 | 0.999 | 1.43 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 0.615 | 3.15 | 1 | 1 | 4.94 | 1 | 1 | 0.986 | 0.989 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.624 | 2.67 | 1 | 1 | 2.72 | 1 | 1 | 7e+03 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.622 | 2.75 | 1 | 1 | 3.07 | 1 | 1 | 6.27e+03 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 0.0874 | 0.309 | 0.756 | 0.913 | 0.368 | 1.32 | 0.126 | 1.17 | 1.26 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.615 | 3.15 | 1 | 0.997 | 4.93 | 1 | 1 | 0.999 | 46.8 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.64 | 2.87 | 1 | 0.946 | 4.21 | 6.52 | 1 | 0.999 | 1.85 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 0.124 | 0.51 | 1.4 | 0.786 | 0.778 | 1.36 | 0.174 | 2.17 | 1.35 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.618 | 2.99 | 1 | 1 | 4.04 | 1 | 1 | 2.79e+03 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.615 | 3.15 | 1 | 1 | 4.94 | 1 | 1 | 0.983 | 0.993 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 0.0776 | 0.325 | 1.01 | 0.993 | 0.528 | 1.02 | 0.102 | 1.28 | 1.09 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.615 | 3.15 | 1 | 0.999 | 4.93 | 1 | 1 | 0.999 | 43.5 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.615 | 3.15 | 1 | 0.998 | 4.94 | 1 | 1 | 0.996 | 0.911 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 0.132 | 0.389 | 0.981 | 1.04 | 0.557 | 1.8 | 0.167 | 1.73 | 1.92 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.615 | 3.15 | 1 | 1 | 4.94 | 1 | 1 | 0.964 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.615 | 3.15 | 1 | 1 | 4.94 | 1 | 1 | 0.988 | 1.02 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 0.219 | 0.369 | 2.06 | 0.265 | 0.472 | 4.04 | 0.245 | 1.75 | 1.6 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.615 | 3.15 | 1 | 0.999 | 4.93 | 1 | 1 | 0.998 | 61.1 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.615 | 3.15 | 1 | 0.988 | 4.94 | 1 | 1 | 0.996 | 0.928 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 69

$T_0$ = 935.0 K  $P_0$ = 2.045 atm  $\phi_0$ = 0.523  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference  Baseline10k (lambda=0)  Soft10k (lambda=1000)  Soft long (lambda=1000)

**1 ANN10**

**2 P-only DeepONet10**

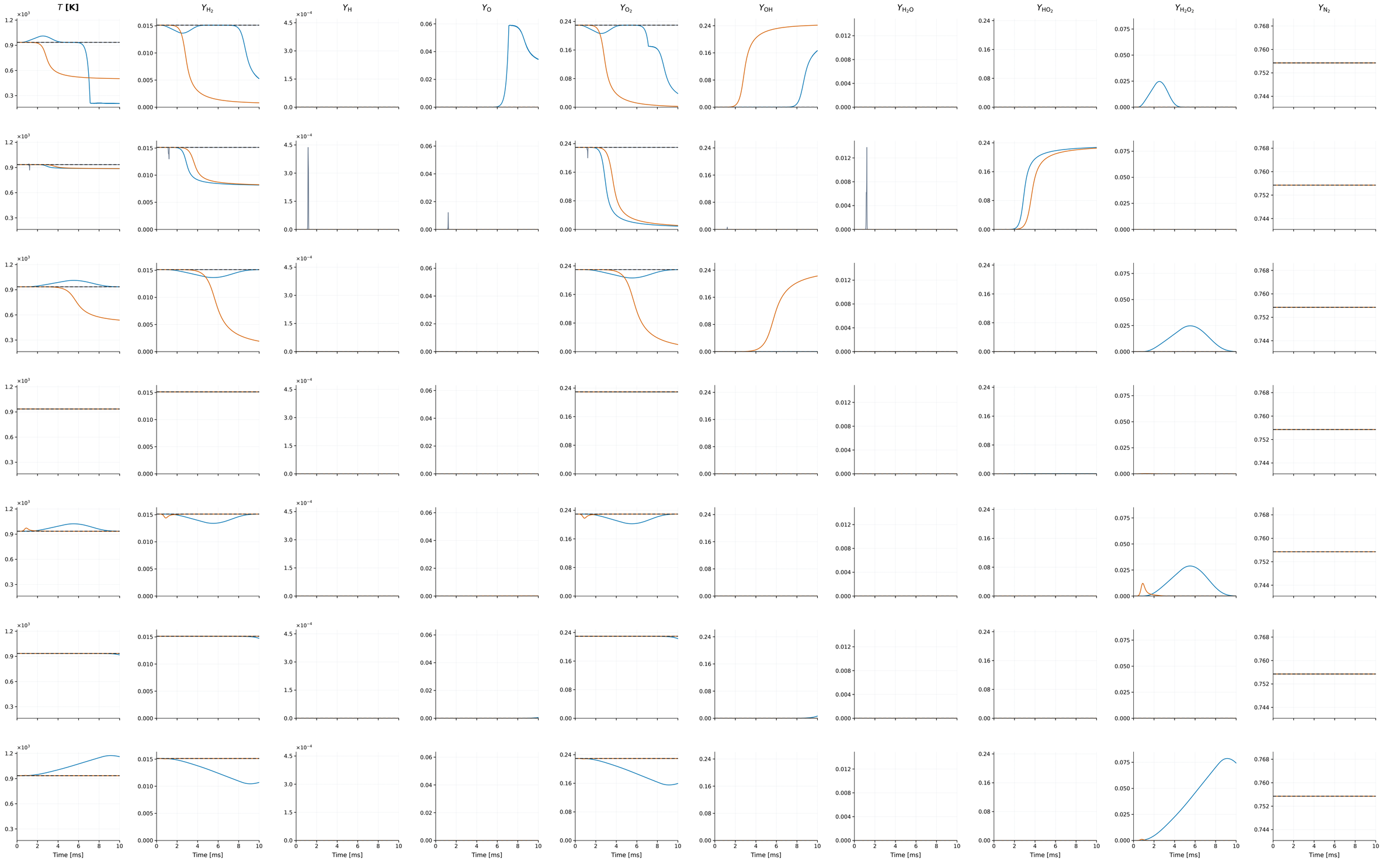
**3 ANN10 + P-only DeepONet10**

**4 T/P DeepONet10**

**5 ANN10 + T/P DeepONet10**

**6 T/P/$\phi$ DeepONet10**

**7 ANN10 + T/P/$\phi$ DeepONet10**




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 69 | Cantera vs NN NRMSE

T0=935.0 K  P0=2.045 atm  phi0=0.523  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

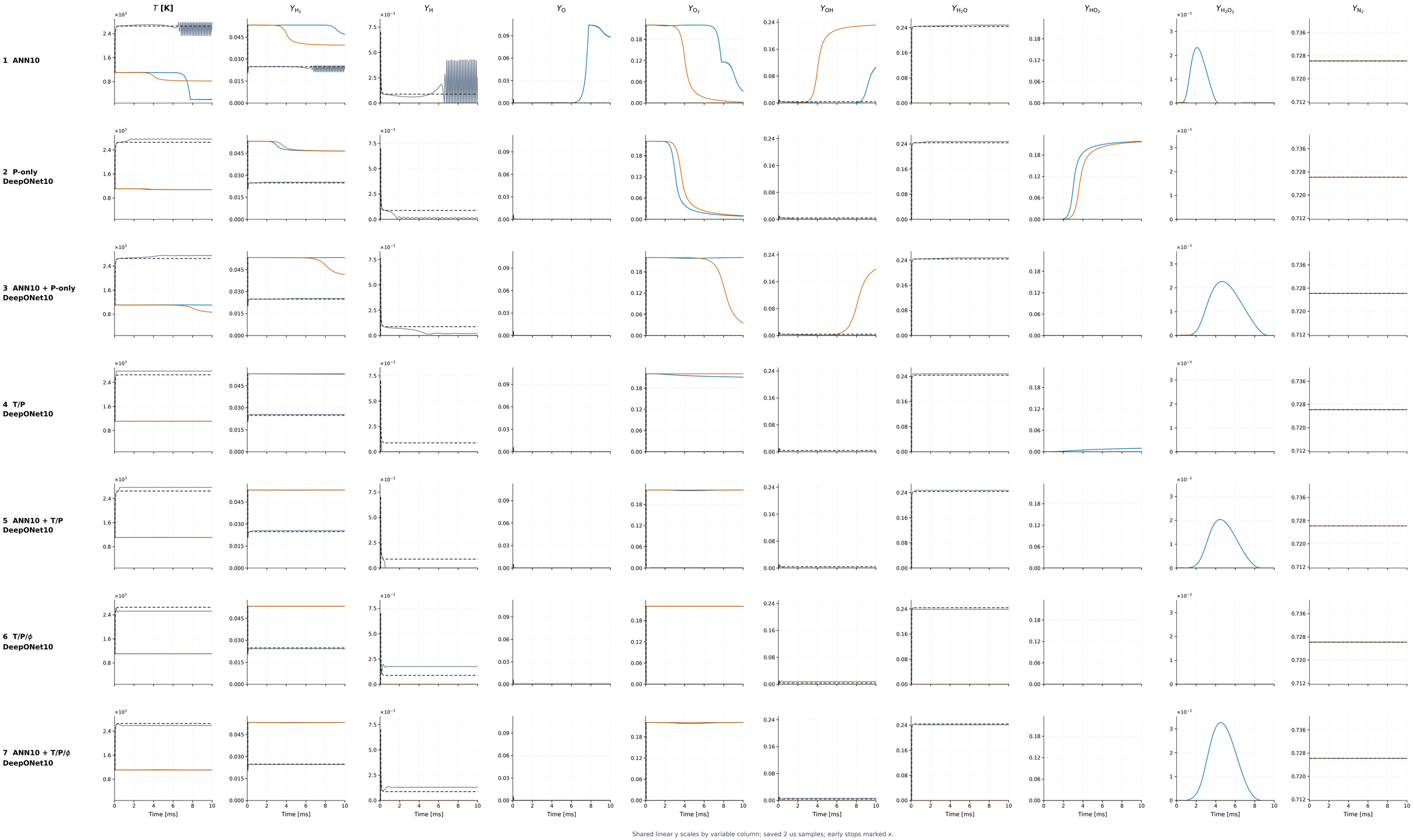
| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | **base10k** | 9.88e-07 | 1.28e-06 | 1 | 185 | 1.53e-06 | 1 | 1 | 0.967 | 0.997 | 0 | Complete |
| **1 ANN10** | **soft10k** | 0.426 | 0.205 | 1 | 2.86e+09 | 0.3 | 1.45e+10 | 1 | 1.12 | 1.06e+05 | 0 | Complete |
| **1 ANN10** | **long** | 0.363 | 0.744 | 1 | 1 | 0.778 | 5.33e+10 | 1 | 1.01 | 132 | 0 | Complete |
| **2 P-only DeepONet10** | **base10k** | 2.83e-03 | 8.45e-03 | 9.87e+06 | 5.45e+07 | 6.11e-03 | 7.94e+07 | 1.15e+04 | 139 | 52.9 | 0 | Complete |
| **2 P-only DeepONet10** | **soft10k** | 0.0405 | 0.361 | 1 | 1 | 0.756 | 1 | 1 | 7.95e+05 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | **long** | 0.0374 | 0.333 | 1 | 1 | 0.697 | 1 | 1 | 7.33e+05 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **base10k** | 9.70e-07 | 1.29e-06 | 1 | 0.492 | 1.55e-06 | 0.998 | 0.999 | 0.988 | 0.999 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **soft10k** | 0.045 | 0.0536 | 1 | 987 | 0.0561 | 3.15e+06 | 1 | 1.23 | 1.56e+05 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | **long** | 0.241 | 0.497 | 1 | 1 | 0.52 | 3.56e+10 | 1 | 1.02 | 154 | 0 | Complete |
| **4 T/P DeepONet10** | **base10k** | 9.45e-07 | 1.27e-06 | 1 | 2.28 | 1.55e-06 | 0.952 | 0.963 | 0.997 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | **soft10k** | 1.08e-04 | 9.71e-04 | 1 | 1 | 2.03e-03 | 1 | 1 | 2.14e+03 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | **long** | 2.47e-04 | 3.35e-04 | 1 | 0.996 | 3.88e-04 | 1 | 0.998 | 81 | 866 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **base10k** | 3.86e-04 | 5.04e-04 | 1 | 13 | 5.70e-04 | 0.999 | 8.88 | 94.2 | 1.33e+03 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **soft10k** | 0.0516 | 0.0616 | 1 | 1.23e+03 | 0.0644 | 2.90e+06 | 1 | 1.45 | 1.79e+05 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | **long** | 7.80e-03 | 9.30e-03 | 1.21e+03 | 7.82e+06 | 9.82e-03 | 4.04e+05 | 0.833 | 252 | 2.68e+04 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **base10k** | 2.78e-05 | 2.32e-05 | 87.3 | 109 | 1.84e-05 | 399 | 35.7 | 7.04 | 8.79 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **soft10k** | 3.08e-03 | 4.29e-03 | 1 | 9.25e+06 | 4.87e-03 | 3.07e+08 | 1 | 103 | 1.29 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | **long** | 1.77e-05 | 2.46e-05 | 0.999 | 48.6 | 3.02e-05 | 1 | 1 | 13.5 | 63.7 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **base10k** | 9.40e-06 | 8.94e-06 | 18.9 | 16.9 | 8.37e-06 | 52.1 | 11.3 | 4.03 | 5.05 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **soft10k** | 0.15 | 0.181 | 1 | 1 | 0.189 | 7.23e+05 | 1 | 3.62 | 5.25e+05 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **long** | 7.57e-04 | 8.00e-04 | 1.90e+04 | 3.48e+06 | 8.96e-04 | 78.3 | 0.974 | 108 | 2.19e+03 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 73

$T_0$ = 1106.0 K  $P_0$ = 1.911 atm  $\phi_0$ = 1.915  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=1000) · Soft long (lambda=1000)




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 73 | Cantera vs NN NRMSE

T0=1106.0 K  P0=1.911 atm  phi0=1.915  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 0.0381 | 0.0487 | 1.14 | 0.997 | 0.157 | 0.82 | 0.0187 | 0.373 | 6.14 | 0 | Complete |
| **1 ANN10** | soft10k | 0.684 | 1.11 | 1 | 199 | 10.9 | 6.27 | 1 | 1 | 1.85e+03 | 0 | Complete |
| **1 ANN10** | long | 0.645 | 0.851 | 1 | 0.999 | 7.61 | 40.7 | 1 | 1 | 15.1 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 0.0376 | 0.025 | 0.757 | 0.734 | 0.154 | 0.77 | 0.019 | 0.37 | 0.648 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.59 | 0.958 | 1 | 1 | 6.65 | 1 | 1 | 1.69e+04 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.589 | 0.976 | 1 | 1 | 7.31 | 1 | 1 | 1.58e+04 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 0.03 | 0.0143 | 0.592 | 0.58 | 7.95e-03 | 0.633 | 0.0124 | 0.0794 | 0.188 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.582 | 1.13 | 1 | 1 | 12.3 | 1 | 1 | 1 | 2.65e+03 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.6 | 1.05 | 1 | 0.998 | 10.9 | 18 | 1 | 1 | 17.4 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 0.0466 | 0.0258 | 0.956 | 0.772 | 0.0161 | 0.986 | 0.0171 | 0.459 | 6.48 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.583 | 1.12 | 1 | 1 | 12 | 1 | 1 | 627 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.582 | 1.13 | 1 | 1 | 12.3 | 1 | 1 | 0.999 | 1.98 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 0.0444 | 0.0276 | 0.93 | 0.673 | 0.0709 | 0.961 | 0.0159 | 0.422 | 2.4 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.582 | 1.13 | 1 | 1 | 12.3 | 1 | 1 | 1 | 2.19e+03 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.582 | 1.13 | 1 | 1 | 12.3 | 1 | 1 | 1 | 5.09 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 0.0497 | 0.0333 | 0.96 | 1.13 | 0.242 | 1.05 | 0.0279 | 0.634 | 0.879 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.582 | 1.13 | 1 | 1 | 12.3 | 1 | 1 | 1.78 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.582 | 1.13 | 1 | 1 | 12.3 | 1 | 1 | 0.999 | 2.91 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 0.0248 | 0.0245 | 0.472 | 0.561 | 0.21 | 0.58 | 0.0195 | 0.528 | 0.672 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.581 | 1.13 | 1 | 1 | 12.3 | 1 | 1 | 1 | 3.49e+03 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.582 | 1.13 | 1 | 0.999 | 12.3 | 1 | 1 | 1 | 3.9 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 74

$T_0$ = 1222.9 K  $P_0$ = 4.971 atm  $\phi_0$ = 1.766  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference  Baseline10k (lambda=0)  Soft10k (lambda=1000)  Soft long (lambda=1000)

**1 ANN10**

**2 P-only DeepONet10**

**3 ANN10 + P-only DeepONet10**

**4 T/P DeepONet10**

**5 ANN10 + T/P DeepONet10**

**6 T/P/$\phi$ DeepONet10**

**7 ANN10 + T/P/$\phi$ DeepONet10**

$T$ [K]  $Y_{H_2}$  $Y_H$  $Y_O$  $Y_{O_2}$  $Y_{OH}$  $Y_{H_2O}$  $Y_{HO_2}$  $Y_{H_2O_2}$  $Y_{N_2}$

Time [ms]

Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 74 | Cantera vs NN NRMSE

T0=1222.9 K  P0=4.971 atm  phi0=1.766  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 0.0331 | 0.0696 | 1.38 | 0.728 | 0.0408 | 0.804 | 0.0201 | 0.211 | 1.99 | 0 | Complete |
| **1 ANN10** | soft10k | 0.681 | 1.33 | 1 | 208 | 23.4 | 4.08 | 1 | 1 | 282 | 0 | Complete |
| **1 ANN10** | long | 0.693 | 1.08 | 1 | 126 | 16.9 | 24.6 | 1 | 1 | 13.3 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 0.0501 | 0.0284 | 1.01 | 1.17 | 0.177 | 1.1 | 0.0287 | 0.692 | 1.41 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.57 | 1.16 | 1 | 1 | 15 | 1 | 1 | 1.93e+04 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.569 | 1.19 | 1 | 1 | 17.2 | 1 | 1 | 1.68e+04 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 0.045 | 0.0562 | 1.06 | 2.06 | 0.212 | 1.67 | 0.0447 | 0.662 | 3.9 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.563 | 1.35 | 1 | 1 | 26.8 | 1 | 1 | 0.999 | 333 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.58 | 1.27 | 1 | 6.08 | 23.9 | 12.5 | 1 | 1 | 16.5 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 0.04 | 0.0532 | 0.97 | 1.09 | 0.72 | 1 | 0.0224 | 1.15 | 9.51 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.563 | 1.35 | 1 | 1 | 26.8 | 1 | 1 | 1.11 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.563 | 1.35 | 1 | 1 | 26.8 | 1 | 1 | 0.998 | 1.39 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 8.40e-03 | 0.0292 | 0.178 | 0.524 | 0.521 | 0.199 | 0.0177 | 1.07 | 0.981 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.563 | 1.35 | 1 | 0.999 | 26.8 | 1 | 1 | 0.999 | 136 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.563 | 1.35 | 1 | 1 | 26.8 | 1 | 1 | 0.998 | 1.99 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 0.155 | 0.0323 | 2.06 | 2 | 0.572 | 5.48 | 0.131 | 1.5 | 5.07 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.563 | 1.35 | 1 | 1 | 26.8 | 1 | 1 | 1.15 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.563 | 1.35 | 1 | 1 | 26.8 | 1 | 1 | 0.998 | 3.4 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 0.152 | 0.0322 | 1.86 | 0.606 | 0.517 | 5.99 | 0.138 | 1.4 | 6.34 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.563 | 1.35 | 1 | 0.996 | 26.8 | 1 | 1 | 0.999 | 496 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.563 | 1.35 | 1 | 1 | 26.8 | 1 | 1 | 0.999 | 2.6 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 76

$T_0$ = 1523.6 K  $P_0$ = 3.626 atm  $\phi_0$ = 1.349  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference  Baseline10k (lambda=0)  Soft10k (lambda=1000)  Soft long (lambda=1000)

**1 ANN10**

**2 P-only DeepONet10**

**3 ANN10 + P-only DeepONet10**

**4 T/P DeepONet10**

**5 ANN10 + T/P DeepONet10**

**6 T/P/$\phi$ DeepONet10**

**7 ANN10 + T/P/$\phi$ DeepONet10**

$T$ [K]  $Y_{H_2}$  $Y_H$  $Y_O$  $Y_{O_2}$  $Y_{OH}$  $Y_{H_2O}$  $Y_{HO_2}$  $Y_{H_2O_2}$  $Y_{N_2}$

Time [ms]

Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 76 | Cantera vs NN NRMSE

T0=1523.6 K  P0=3.626 atm  phi0=1.349  |  Repaired self-fed,10ms

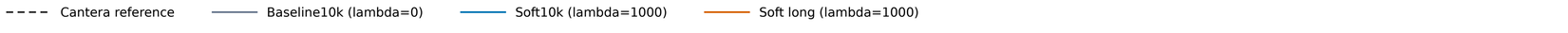
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 7.28e-03 | 0.157 | 0.733 | 1.66 | 0.164 | 0.967 | 0.0661 | 3.08 | 10.3 | 0 | Complete |
| **1 ANN10** | soft10k | 0.641 | 2.57 | 1 | 26.9 | 28.6 | 1.27 | 1 | 0.971 | 2.03e+03 | 0 | Complete |
| **1 ANN10** | long | 0.56 | 1.81 | 1 | 0.999 | 17.6 | 9.73 | 1 | 0.995 | 106 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 0.0606 | 0.124 | 0.537 | 0.631 | 0.585 | 0.637 | 0.075 | 1.43 | 3.56 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.498 | 2.24 | 1 | 1 | 19 | 1 | 1 | 2.53e+04 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.498 | 2.3 | 1 | 1 | 21.3 | 1 | 1 | 2.27e+04 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 0.0447 | 0.119 | 0.469 | 0.535 | 0.487 | 0.537 | 0.062 | 1.43 | 4.32 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.491 | 2.61 | 1 | 1 | 33.3 | 1 | 1 | 0.927 | 2.1e+03 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.528 | 2.21 | 1 | 0.998 | 25.1 | 6.29 | 1 | 0.99 | 135 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 0.101 | 0.138 | 1.02 | 0.564 | 0.519 | 1.13 | 0.102 | 4.47 | 2.83 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.492 | 2.62 | 1 | 1 | 33.5 | 1 | 1 | 1.56 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.492 | 2.62 | 1 | 1 | 33.5 | 1 | 1 | 0.873 | 0.854 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 0.0593 | 0.113 | 0.713 | 0.206 | 0.165 | 0.838 | 0.067 | 0.92 | 1.56 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.491 | 2.61 | 1 | 1 | 33.3 | 1 | 1 | 0.902 | 2.03e+03 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.492 | 2.62 | 1 | 1 | 33.5 | 1 | 1 | 0.783 | 6.46 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 0.0794 | 0.0678 | 0.632 | 0.485 | 0.728 | 0.848 | 0.0779 | 2.08 | 2.75 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.492 | 2.62 | 1 | 1 | 33.5 | 1 | 1 | 0.529 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.492 | 2.62 | 1 | 1 | 33.5 | 1 | 1 | 0.864 | 3.35 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 0.0672 | 0.0325 | 0.52 | 0.132 | 0.106 | 0.894 | 0.0693 | 0.916 | 1.18 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.491 | 2.61 | 1 | 1 | 33.3 | 1 | 1 | 0.899 | 2.4e+03 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.492 | 2.62 | 1 | 0.997 | 33.5 | 1 | 1 | 3.32 | 54.6 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 78

$T_0$ = 962.3 K  $P_0$ = 2.443 atm  $\phi_0$ = 1.140  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$




**1 ANN10**

**2 P-only DeepONet10**

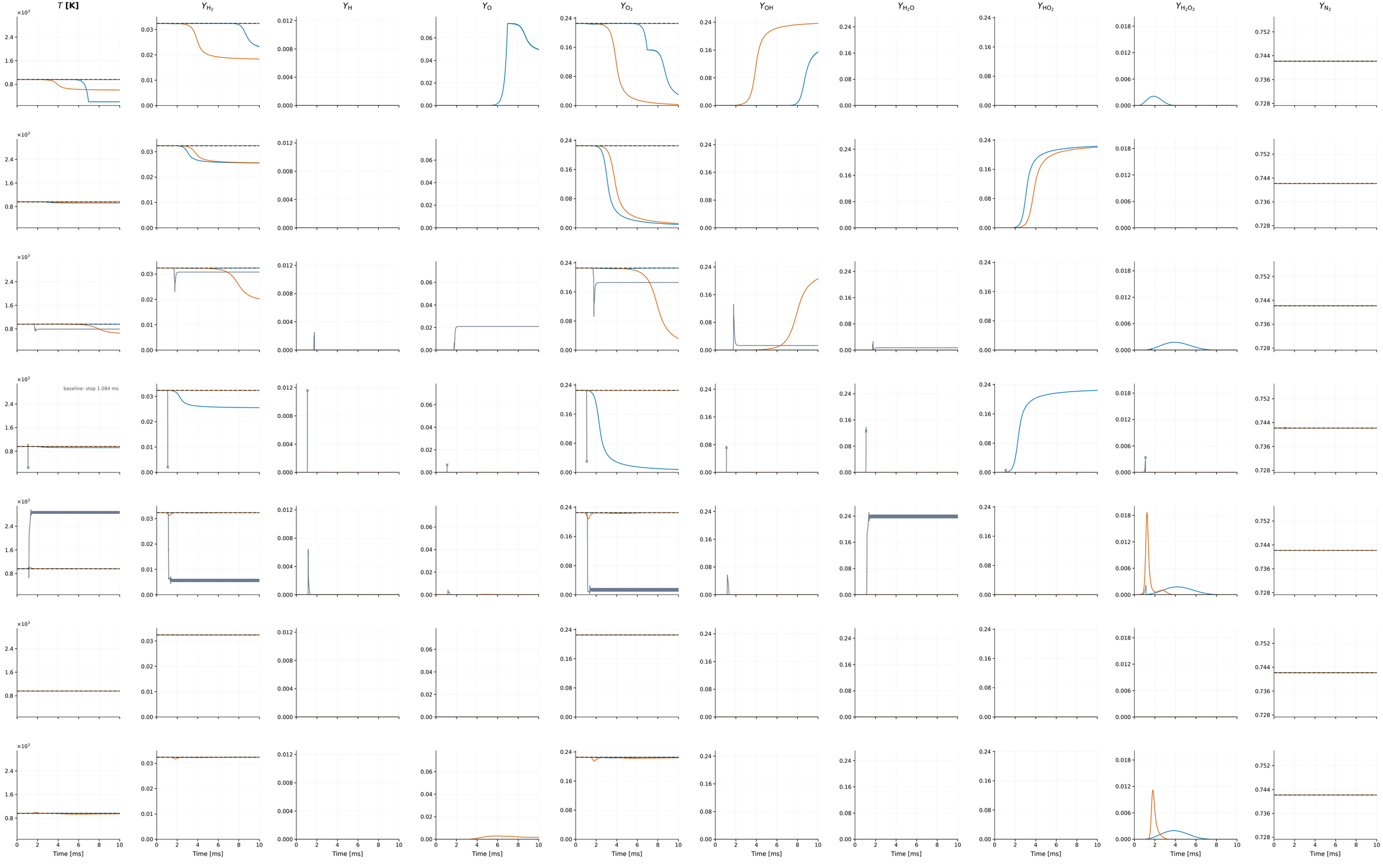
**3 ANN10 + P-only DeepONet10**

**4 T/P DeepONet10**

**5 ANN10 + T/P DeepONet10**

**6 T/P/$\phi$ DeepONet10**

**7 ANN10 + T/P/$\phi$ DeepONet10**




Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 78 | Cantera vs NN NRMSE

T0=962.3 K  P0=2.443 atm  phi0=1.140  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 4.72e-05 | 2.63e-05 | 3.24 | 4.29e+03 | 6.52e-05 | 1 | 1 | 1.07 | 2.06 | 0 | Complete |
| **1 ANN10** | soft10k | 0.441 | 0.088 | 1 | 1.27e+08 | 0.324 | 3.33e+08 | 1 | 1 | 160 | 0 | Complete |
| **1 ANN10** | long | 0.268 | 0.316 | 1 | 1.2e+03 | 0.721 | 1.20e+09 | 1 | 1 | 1.75 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 3.54e-05 | 1.95e-05 | 1 | 0.876 | 4.00e-05 | 0.999 | 0.869 | 0.848 | 0.718 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.0319 | 0.164 | 1 | 1 | 0.746 | 1 | 1 | 3.68e+04 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.0294 | 0.151 | 1 | 1 | 0.687 | 1 | 1 | 3.40e+04 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 0.16 | 0.0496 | 8.47e+05 | 6.63e+07 | 0.166 | 1.05e+08 | 1.77e+03 | 16.5 | 4.06 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 2.37e-03 | 1.60e-03 | 1 | 1.23e+04 | 3.65e-03 | 3.17e+03 | 1 | 1 | 188 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.127 | 0.151 | 1 | 2.32e+03 | 0.344 | 5.71e+08 | 1 | 1 | 1.93 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | — | — | — | — | — | — | — | — | — | — | Stop 1.084 ms |
| **4 T/P DeepONet10** | soft10k | 0.0339 | 0.174 | 1 | 1 | 0.793 | 1 | 1 | 3.92e+04 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 4.00e-05 | 2.23e-05 | 1 | 1 | 4.62e-05 | 1 | 1 | 0.882 | 1.13 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 1.86 | 0.777 | 1.64e+06 | 1.13e+06 | 0.886 | 3.31e+07 | 5.98e+04 | 8.67 | 37.4 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 2.40e-03 | 1.62e-03 | 1 | 2.14e+03 | 3.69e-03 | 2.04e+03 | 1 | 1 | 191 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 7.83e-03 | 5.28e-03 | 256 | 3.55e+05 | 0.0121 | 5.22e+04 | 0.999 | 14.8 | 616 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 4.54e-05 | 2.59e-05 | 1 | 0.884 | 5.42e-05 | 0.997 | 1 | 1 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 4.85e-05 | 2.03e-05 | 1 | 1 | 8.15e-05 | 1 | 1 | 4.69 | 1 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 4.10e-05 | 2.26e-05 | 1 | 0.998 | 4.62e-05 | 1 | 1 | 0.934 | 0.771 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 4.54e-05 | 2.58e-05 | 1 | 0.819 | 5.42e-05 | 0.995 | 0.999 | 0.999 | 1 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 2.54e-03 | 1.71e-03 | 1 | 2.06e+04 | 3.90e-03 | 2.45e+03 | 1 | 1 | 201 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.0192 | 3.60e-03 | 3.93e+03 | 6.24e+06 | 0.0114 | 5.48e+04 | 1 | 10.4 | 419 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

# Lambda 1000 continuation: Cantera vs NN | Test IC 88

$T_0$ = 1502.1 K  $P_0$ = 4.980 atm  $\phi_0$ = 0.859  |  Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference  Baseline10k (lambda=0)  Soft10k (lambda=1000)  Soft long (lambda=1000)

**1 ANN10**

**2 P-only DeepONet10**

**3 ANN10 + P-only DeepONet10**

**4 T/P DeepONet10**

**5 ANN10 + T/P DeepONet10**

**6 T/P/$\phi$ DeepONet10**

**7 ANN10 + T/P/$\phi$ DeepONet10**

$T$ [K]  $Y_{H_2}$  $Y_H$  $Y_O$  $Y_{O_2}$  $Y_{OH}$  $Y_{H_2O}$  $Y_{HO_2}$  $Y_{H_2O_2}$  $Y_{N_2}$

Time [ms]

Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

# Lambda 1000 | Test IC 88 | Cantera vs NN NRMSE

T0=1502.1 K  P0=4.980 atm  phi0=0.859  |  Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | base10k | 0.0536 | 0.764 | 5.02 | 0.609 | 0.392 | 0.914 | 0.0409 | 1.27 | 4.48 | 0 | Complete |
| **1 ANN10** | soft10k | 0.658 | 10.3 | 1 | 13.1 | 4.24 | 1.42 | 1 | 0.981 | 509 | 0 | Complete |
| **1 ANN10** | long | 0.567 | 6.91 | 1 | 0.992 | 2.81 | 7.86 | 1 | 0.998 | 35.7 | 0 | Complete |
| **2 P-only DeepONet10** | base10k | 0.101 | 0.92 | 0.803 | 0.75 | 0.101 | 1.19 | 0.157 | 5.76 | 17.3 | 0 | Complete |
| **2 P-only DeepONet10** | soft10k | 0.498 | 8.64 | 1 | 1 | 2.81 | 1 | 1 | 6.21e+03 | 1 | 0 | Complete |
| **2 P-only DeepONet10** | long | 0.496 | 9.03 | 1 | 1 | 3.2 | 1 | 1 | 5.38e+03 | 1 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | base10k | 0.234 | 0.143 | 2.9 | 0.928 | 0.832 | 4.28 | 0.3 | 1.4 | 7.8 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | soft10k | 0.489 | 10.6 | 1 | 1 | 5.07 | 1 | 1 | 0.953 | 625 | 0 | Complete |
| **3 ANN10 + P-only DeepONet10** | long | 0.526 | 8.92 | 1 | 0.985 | 3.92 | 4.65 | 1 | 0.996 | 44.9 | 0 | Complete |
| **4 T/P DeepONet10** | base10k | 0.0871 | 0.306 | 1.15 | 0.406 | 0.106 | 0.998 | 0.107 | 2.11 | 2.76 | 0 | Complete |
| **4 T/P DeepONet10** | soft10k | 0.49 | 10.6 | 1 | 1 | 5.09 | 1 | 1 | 0.908 | 1 | 0 | Complete |
| **4 T/P DeepONet10** | long | 0.49 | 10.6 | 1 | 1 | 5.09 | 1 | 1 | 0.919 | 0.675 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | base10k | 0.0113 | 0.0567 | 0.186 | 0.0697 | 0.0111 | 0.105 | 0.0136 | 0.111 | 0.106 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | soft10k | 0.489 | 10.6 | 1 | 1 | 5.07 | 1 | 1 | 0.879 | 496 | 0 | Complete |
| **5 ANN10 + T/P DeepONet10** | long | 0.49 | 10.6 | 1 | 1 | 5.09 | 1 | 1 | 0.885 | 2.78 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | base10k | 0.024 | 0.107 | 0.479 | 0.124 | 0.0122 | 0.217 | 0.0275 | 0.295 | 0.232 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | soft10k | 0.49 | 10.6 | 1 | 1 | 5.09 | 1 | 1 | 0.833 | 0.997 | 0 | Complete |
| **6 T/P/$\phi$ DeepONet10** | long | 0.49 | 10.6 | 1 | 1 | 5.09 | 1 | 1 | 0.932 | 0.982 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | base10k | 0.0144 | 0.076 | 0.19 | 0.0787 | 0.012 | 0.145 | 0.0184 | 0.122 | 0.129 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | soft10k | 0.491 | 10.6 | 1 | 0.981 | 5.06 | 1 | 1 | 0.86 | 554 | 0 | Complete |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | long | 0.49 | 10.6 | 1 | 1 | 5.09 | 1 | 1 | 0.938 | 3.8 | 0 | Complete |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.